Genetics and Genomics

The BTB-ZF gene *Bm-mamo* regulates pigmentation in silkworm caterpillars

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Abstract

The color pattern of insects is one of the most dazzling adaptive evolutionary phenotypes. However, the molecular regulation of this color pattern is not fully understood. In this study, we found that the transcription factor Bm-mamo is responsible for *black dilute* (*bd*) allele mutations in the silkworm. Bm-mamo belongs to the BTB zinc finger family and is orthologous to mamo in *Drosophila melanogaster*. This gene has a conserved function in gamete production in *Drosophila* and silkworms and evolved a pleiotropic function in the regulation of color patterns in caterpillars. We found that *Bm-mamo* can comprehensively regulate the expression of related pigment synthesis and cuticular protein genes to form color patterns. This indicates that insects have a genetic basis for coordinate regulation of the structure and shape of the cuticle, as well as color patterns. This genetic basis provides the possibility for constructing the complex appearances of some insects. This study provides new insight into the regulation of color patterns.

Teaser

The color patterns of insects are highly exquisite and significantly divergent. The metabolism of pigments is the material basis for insect coloration. However, the cuticle of insects plays an important role as a scaffold for carrying pigment particles. Cuticular proteins are some of the main components of the cuticle. This study showed that a BTB-ZF family transcription factor protein, Bm-mamo, can comprehensively regulate melanin synthesis and the expression of multiple cuticular protein genes. Hence, insects have a genetic basis for integrated control of cuticle and color pattern construction, which enables them to produce complex appearances.

eLife assessment

This **important** study identifies the gene mamo as a new regulator of pigmentation in the silkworm Bombyx mori, a function that was previously unsuspected based on extensive work on Drosophila where the mamo gene is involved in gamete production. The evidence supporting the role of Bm-nano in pigmentation is **convincing**, including high-resolution linkage mapping of two mutant strains, expression profiling, and reproduction of the mutant phenotypes with state-of-the-art RNAi and CRISPR knock-out assays. While the discussion about genetic changes being guided or accelerated by the environment is extremely speculative and has little relevance for the findings presented, the work will be of interest to evolutionary biologists and geneticists studying color patterns and evolution of gene networks.

Introduction

Insects often display stunning colors, and the patterns of these colors have been shown to be involved in behavior(*1*); immunity(*2*); thermoregulation(*3*); UV protection(*4*); and especially visual antagonism of predators via aposematism(*5*), mimicry(*6*), cryptic color patterns(*7*) or some combination of the above (*8*). In addition, the color patterns are divergent and convergent among populations(*9*). Due to these striking visual features and highly active adaptive evolutionary phenotypes, the genetic basis and evolutionary mechanism of color patterns have been a topic of interest for a long time.

Insect coloration can be pigmentary, structural, or bioluminescent. Pigments are mainly synthesized by the insects themselves and form solid particles that are deposited in the cuticle of the body surface and the scales of the wings(*10*, *11*). Interestingly, recent studies have found that bile pigments and carotenoid pigments synthesized through biological synthesis are incorporated into body fluids and filled in the wing membranes of two butterflies (*Siproeta stelenes* and *Philaethria diatonica*) via hemolymph circulation, providing color in the form of liquid pigments(*12*). The pigments form colors by selective absorption and/or scattering of light depending on their physical properties(*13*). However, structural color refers to colors, such as metallic colors and iridescence, generated by optical interference and grating diffraction of the microstructure/nanostructure of the body surface or appendages (such as scales)(*14*, *15*). Pigment color and structural color are widely distributed in insects and can only be observed by the naked eye in illuminated environments. However, some insects, such as fireflies, exhibit colors (green to orange) in the dark due to bioluminescence(*16*). Bioluminescence occurs when luciferase catalyzes the oxidation of small molecules of luciferin(*17*). In conclusion, the color patterns of insects have evolved to be highly sophisticated and are closely related to their living environments. For example, cryptic color can deceive animals via high similarity to the surrounding environment. However, the molecular mechanism by which insects form precise color patterns to match their living environment is still unknown.

Recent research has identified the metabolic pathways of related pigments, such as melanins, pterins, and ommochromes, in Lepidoptera(*18*). A deficiency of enzymes in the pigment metabolism pathway can lead to changes in color patterns(*19*, *20*). In addition to pigment synthesis, the microstructure/nanostructure of the body surface and wing scales are important factors that influence body color patterns. The body surface and wing scales of Lepidoptera are mainly composed of cuticular proteins (CPs) (*21*, *22*). There are multiple genes encoding CPs in Lepidopteran genomes. For example, in *Bombyx mori,* over 220 genes encode CPs(23). However, there are still many unknowns regarding the function of CPs and the molecular mechanisms underlying their fine-scale localization in the cuticle.

In addition, some pleiotropic factors, such as *wnt1*(24), *Apontic-like*(25), *clawless*(26), *abdominal-A*(27), *abdominal-B*(28), *engrailed*(29), *antennapedia*(30), *optix*(31), *bric à brac* (*bab*)(32), and *Distal-less* (*Dll*)(33), play an important role in the regulation of color patterns. The molecular mechanism by which these factors participate in the regulation of body color patterns needs further study. In addition, there are many undiscovered factors in the gene regulatory network (GRN) involved in the regulation of insect color patterns. The identification of new color pattern regulatory genes and the study of their molecular mechanisms would be helpful for further understanding color patterns.

Silkworms (*B. mori*) have been completely domesticated for more than 5000 years and are famous for their silk fiber production(34). Because of this long-term domestication, artificial selection and genetic research, its genetic background is well resolved, and approximately 200 strains of mutants with variable color patterns have been preserved(35), which provide a good resource for color pattern research. The *black dilute* (*bd*) mutant, which exhibits recessive Mendelian inheritance, has a dark gray larval body color, and the female is sterile. bd^f (*black dilute fertile*) is one of the alleles, leading to a lighter body color than that of *bd* and fertile females (**Fig. 1**). The two mutations were mapped to a single locus at 22.9 centimorgans (cM) of linkage group 9(36). No pigmentation-related genes have been reported at this locus. Thus, research may reveal a new color pattern-related gene, which stimulated our interest.

Results

Candidate gene of the *bd* allele

To identify the genomic region responsible for the *bd* alleles, positional cloning of the *bd* locus was performed. Due to the female infertility of the *bd* mutant and the fertility of females with the bd^f allele, we used bd^f and the wild-type Dazao strain as the parents for mapping analysis. The 1162 back-crossed filial 1st (BC1M) generation individuals from bd^f and Dazao were used for fine mapping with molecular markers (**Fig. 2**). A genomic region of approximately 390 kilobases (kb) was responsible for the *bd* phenotype (Table S1). In the SilkDB database(37), this region included five predicted genes (*BGIBMGA012516*, *BGIBMGA012517*, *BGIBMGA012518*, *BGIBMGA012519* and *BGIBMGA014089*). In addition, we analyzed the predictive genes for this genomic region from the GenBank(38) and SilkBase(39) databases (Fig. S1). The number of predicted genes varied among different databases. We performed sequence alignment analysis of the predicted genes in the three databases to determine their correspondence. Then, Real-time Quantitative polymerase chain reaction (qPCR) was performed, which showed that *BGIBMGA012517* and *BGIBMGA012518* were significantly downregulated on Day 3 of the fifth instar of larvae in the *bd* phenotype individuals, while there was no difference in the expression levels of other genes (Fig. S1). These two genes were predicted as a single locus (*LOC101738295*) in GenBank. To determine the gene structure of *BGIBMGA012517* and *BGIBMGA012518*, we used forward primers for the *BGIBMGA012517* gene and reverse primers for the *BGIBMGA012518* gene to amplify cDNA from the wild-type Dazao strain. By gene cloning, the predicted genes *BGIBMGA012517* and *BGIBMGA012518* were proven to be one gene. For the convenience of description, we have temporarily called it the *12517/8* gene.

The *12517/8* gene produces two transcripts; the open reading frame (ORF) of the long transcript is 2397 bp, and the ORF of the short transcript is 1824 bp, which shares the same 5'-terminus as the wild-type Dazao strain (Fig. S2). The *12517/8* gene showed significantly lower expression in the bd^f mutant (Fig. S3), and multiple variations were found in the nearby region of this gene by

Fig. 1.

Phenotypes of *bd*, *bd*^{f}, and wild-type Dazao larvae.

The epidermis of *bd* is dark gray, and the epidermis of *bd*^{f} is light gray at the 5th instar (day 3) of silkworm larvae. The bar indicates 1 centimeter.

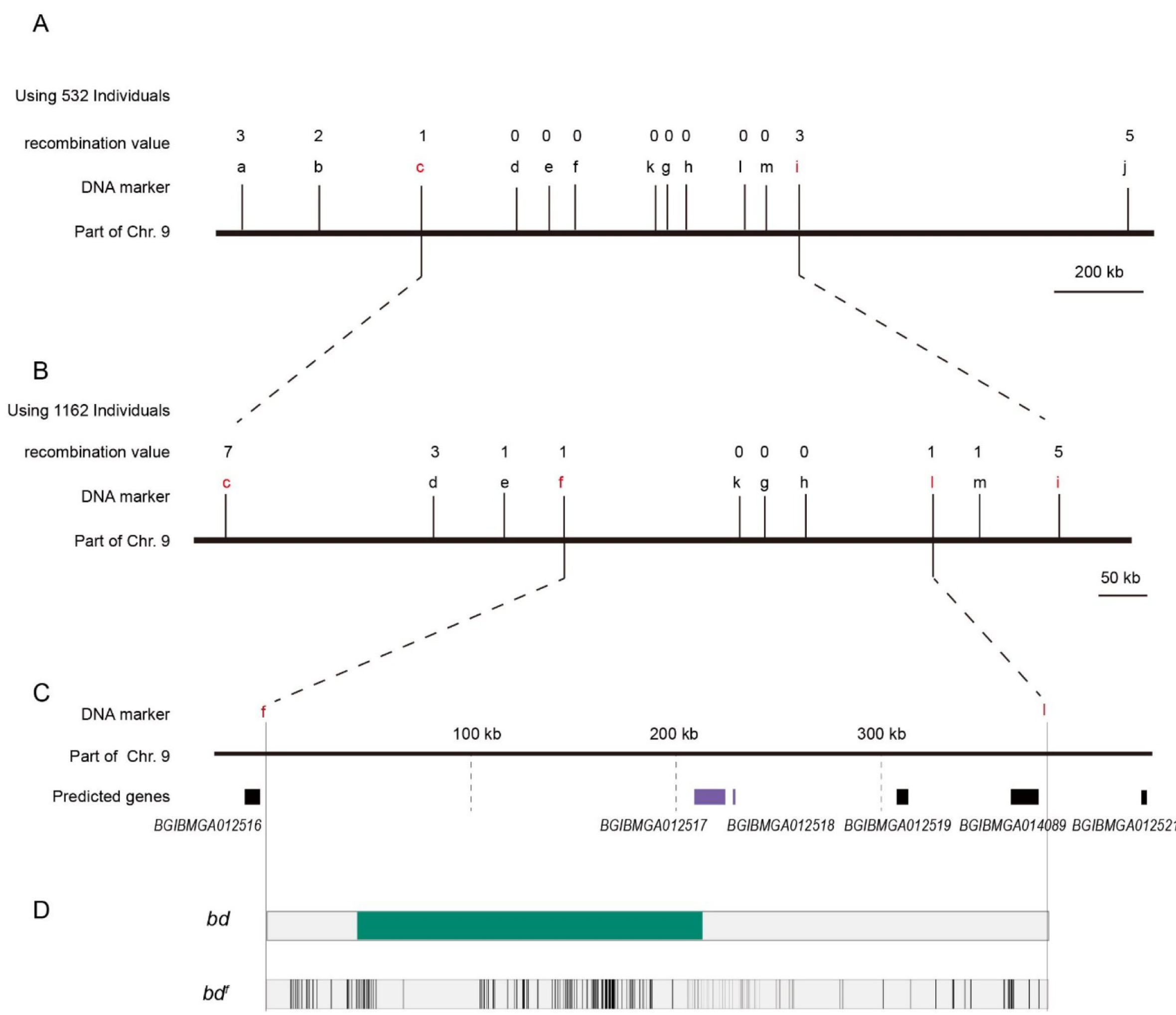


Fig. 2.

Positional cloning of the *bd* locus.

(A) We used 532 BC1 individuals to map the *bd* locus between PCR markers c and i. The numbers above the DNA markers indicate recombination events.
(B) A total of 1162 BC1 individuals were used to narrow the *bd* locus to an approximately 400 kb genomic region.
(C) Partial enlarged view of the region responsible for *bd*. This region contains 4 predicted genes, *BGIBMGA012517*, *BGIBMGA012518*, *BGIBMGA012519* and *BGIBMGA014089*.
(D) Analysis of nucleotide differences in the region responsible for *bd*. The green block indicates the deletion of the genome in *bd* mutants. The black vertical lines indicate the SNPs and indels of *bd^f* mutants.

comparative genomic analysis between bd^f and Dazao (Table S2). In addition, *12517/8* was completely silenced due to the deletion of DNA fragments (approximately 168 kb) from the first upstream intron and an insertion of 3560 bp in the *bd* mutant (Fig. S4).

To predict the function of the *12517*/8 gene, we performed a BLAST search using its full-length sequence and found a transcription factor, the *maternal gene required for meiosis* (*mamo*) in *D. melanogaster*, that had a high sequence similarity to that of the *12517/8* gene (Fig. S5). Therefore, we named the *12517/18* gene *Bm-mamo*; the long transcript was designated as *Bm-mamo-L*, and the short transcript was designated as *Bm-mamo-S*.

The *mamo* gene belongs to the Broad-complex, Tramtrack and bric à brac/poxvirus zinc finger protein (BTB-ZF) family. In the BTB-ZF family, the zinc finger serves as a recognition motif for DNA-specific sequences, and the BTB domain promotes oligomerization and the recruitment of other regulatory factors(*40*). Most of these factors are typically transcription repressors, such as nuclear receptor corepressor (N-CoR) and silencing mediator for retinoid and thyroid hormone receptor (SMRT)(*41*), but some are activators, such as p300(*42*). Therefore, these features commonly serve as regulators of gene expression. *mamo* is enriched in embryonic primordial germ cells (PGCs) in *D. melanogaster*. Individuals deficient in *mamo* are able to undergo oogenesis but fail to execute meiosis properly, leading to female infertility in *D. melanogaster*(*43*). *Bm-mamo* was identified as an important candidate gene for further analysis.

Expression pattern analysis of *Bm-mamo*

To analyze the expression profiles of *Bm-mamo*, we performed quantitative PCR. The expression levels of the *Bm-mamo* gene were investigated in the whole body at different developmental stages, from the embryonic stage to the adult stage, in the Dazao strain. The gene was highly expressed in the molting stage of caterpillars, and its expression was upregulated in the later pupal and adult stages (**Fig. 3A**). This suggests that the *Bm-mamo* gene responds to the ecdysone titer and participates in the processes of molting and metamorphosis in silkworms. In the investigation of tissue-specific expression levels in the 5th-instar 3rd-day larvae of the Dazao strain, the midgut, head, and epidermis had high expression levels; the trachea, nerves, silk glands, testis, ovary, muscle, wing disc and Malpighian tubules had moderate expression levels; and the blood and fat bodies had low expression levels (**Fig. 3B**). This suggests that *Bm-mamo* is involved in the developmental regulation of multiple silkworm tissues. Due to the melanism of the epidermis of the *bd* mutant and the high expression level of the *Bm-mamo* gene in the epidermis, we measured the expression level of this gene in the epidermis of the 4th to 5th instar of the Dazao strain. In the epidermis, the *Bm-mamo* gene was upregulated in the molting period, and the highest expression was observed at the beginning of molting (**Fig. 3C**).

Functional analyses of *Bm-mamo*

To study the function of the *Bm-mamo* gene, we carried out an RNA interference (RNAi) experiment. Short interfering RNA (siRNA) was injected into the hemolymph of silkworms, and an electroporation experiment(*44*) was immediately conducted. We found significant melanin pigmentation in the epidermis of the newly molted 5th-instar larvae. This indicates that *Bm-mamo* deficiency can cause melanin pigmentation (**Fig. 4**). The melanism phenotype of RNAi individuals was similar to that of *bd* mutants.

In addition, gene knockout was performed. Ribonucleoproteins (RNPs) generated by guide RNA (gRNA) and recombinant Cas9 protein were injected into 450 silkworm embryos. In the G0 generation, individuals with a mosaic melanization phenotype were found. These melanistic individuals were raised to moths and then crossed. The homozygous line with gene knockout was obtained through generations of screening. The gene-edited individuals had a significantly melanistic body color, and the female moths were sterile (**Fig. 5**).

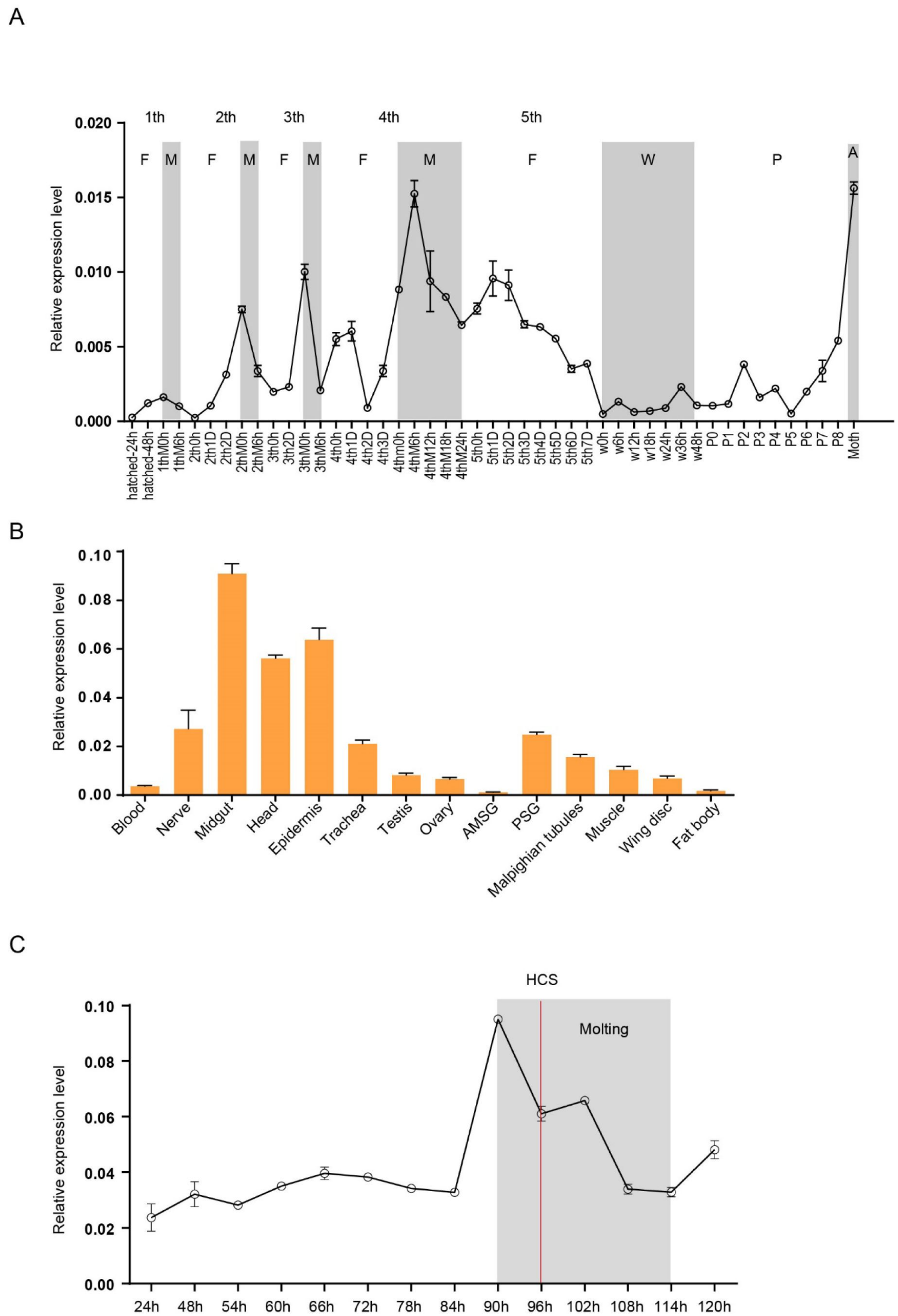


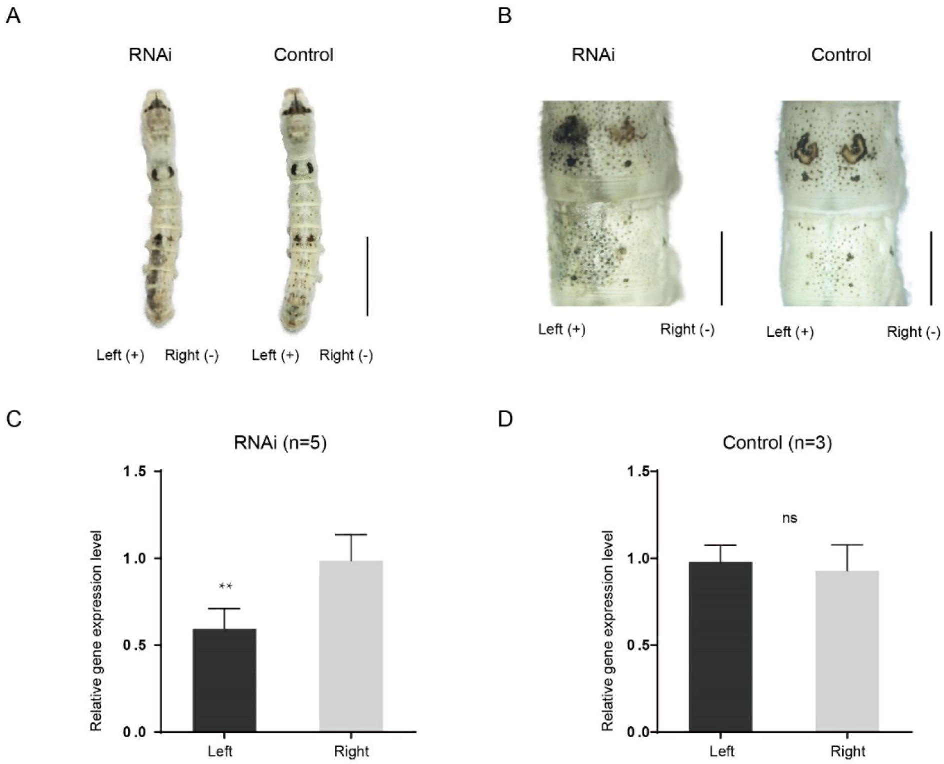
Fig. 3.

Spatiotemporal expression of *Bm-mamo*.

(A) Temporal expression of *Bm-mamo*. In the molting stage and adult stage, this gene is significantly upregulated. M: molting stage, W: wandering stage, P: pupal stage, P1: Day 1 of pupal stage, A: adult stage, 4th3d indicates 4th instar day 3. 1st to 5th denote the first instar of larvae to fifth instar of larvae, respectively.

(B) Tissue-specific expression of 4th-instar molting larvae. *Bm-mamo* had relatively high expression levels in the midgut, head, and epidermis. AMSG: anterior division of silk gland and middle silk gland, PSG: posterior silk gland.

(C) Detailed analysis of *Bm-mamo* at the 4th larval stage in the epidermis of the Dazao strain. *Bm-mamo* expression is upregulated during the molting stage. HCS indicates the head capsule stage. The “h” indicates the hour, 90 h: at 90 hours of the 4th instar.



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	Injected individuals	Melanism phenotype	Survival number	Efficiency
RNAi	48	23	48	47.92%
Control	21	0	17	0

Fig. 4.

(A) siRNA was introduced by microinjection followed by electroporation. "+" and "-" indicate the positive and negative poles of the electrical current. Scale bars, 1 cm.

(B) Partial magnification of the siRNA experimental group and negative control group. Scale bars, 0.2 cm.

(C) and (**D**) Relative expression levels of *Bm-mamo* in the negative control and RNAi groups were determined by qPCR analysis. The means ± s.d.s. Numbers of samples are shown in the upper right in each graph. ***P*<0.01, paired Student's t test (NS, not significant).

(E) Efficiency statistics of RNAi.

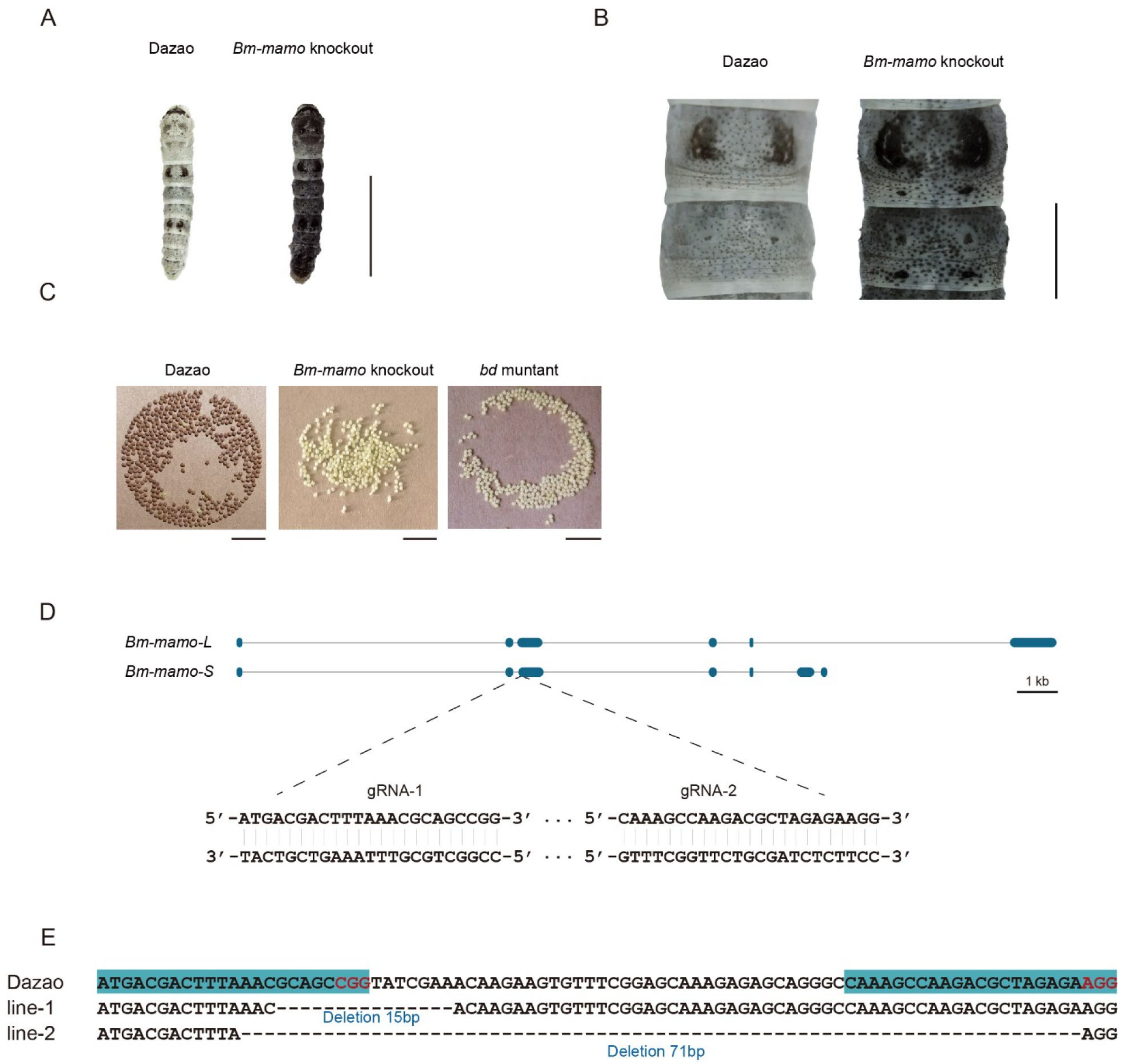

Fig. 5.

Knockout of *Bm-mamo*

(A) Larval phenotype in G3 fourth-instar larvae of Dazao targeted for *Bm-mamo*. Scale bars, 1 cm.
(B) Partial magnification of the *Bm-mamo* knockout individual and control. Scale bars, 0.2 cm.
(C) After 48 hours of laying eggs, the pigmentation of the eggs indicates that those produced by knockout homozygous females cannot undergo normal pigmentation and development.
(D) Genomic structure of *Bm-mamo*. Open reading frame (blue), untranslated region (black). The gRNA 1 and gRNA 2 sequences are shown.
(E) Sequences of the *Bm-mamo* knockout individuals. Lines 1 and 2 indicate deletions of 15 and 71 bp, respectively.

These results indicated that the *Bm-mamo* gene negatively regulates melanin pigmentation in caterpillars and participates in the reproductive regulation of female moths.

Downstream target gene analysis

Bm-mamo belongs to the zinc finger protein family, which specifically recognizes downstream DNA sequences according to their zinc fingers. We identified homologous genes of *mamo* in multiple species and conducted a phylogenetic analysis (Fig. S6). Sequence alignment revealed that the amino acid residues of the zinc finger motif of the mamo-S protein were highly conserved among 57 species. The insufficient prediction of alternative transcripts of mamo in different species in GenBank may have resulted in the mamo-L protein being found in only 30 species (Fig. S7). As the mamo protein carries a tandem Cys_2His_2 zinc finger (C2H2-ZF) motif, it can directly bind to DNA sequences. Previous research has suggested that the ZF-DNA binding interface can be understood as a "canonical binding model", in which each finger contacts DNA in an antiparallel manner. The binding sequence of the C2H2-ZF motif is determined by the amino acid residue sequence of its α-helical component. Considering the first amino acid residue in the α-helical region of the C2H2-ZF domain as position 1, positions –1, 2, 3, and 6 are key amino acids in the recognition and binding of DNA. The residues at positions –1, 3, and 6 specifically interact with base 3, base 2, and base 1 of the DNA sense sequence, respectively, while the residue at position 2 interacts with the complementary DNA strand(*45*, *46*). To analyze the downstream target genes of mamo, we first predicted its DNA binding motifs using online software (*http://zf.princeton.edu*) based on the canonical binding model(*47*). In addition, the DNA-binding sequence of mamo (TGCGT) in *Drosophila*, confirmed by electrophoretic mobility shift assay (EMSA) experiments(*48*), has a consensus sequence with the predicted binding site sequence of Bm-mamo-S (GTGCGTGGC), and the predicted sequence was longer. This indicates that the predicted results of the DNA-binding site have good reliability.

Furthermore, the predicted DNA-binding sites of Bm-mamo-L and Bm-mamo-S were highly consistent with those of mamo orthologs in different species (Fig. S8). This suggests that the protein may regulate similar target genes between species.

C2H2-ZF transcription factors function by recognizing and binding to *cis*-regulatory sequences in the genome, which harbor *cis*-regulatory elements (CREs)(*49*). CREs are broadly classified as promoters, enhancers, silencers and insulators(*50*). CREs are often located near their target genes, such as enhancers, which are typically located upstream (5') or downstream (3') of the gene they regulate, or in introns, but approximately 12% of CREs are located far from their target gene(*51*). Therefore, we first investigated the genome range 2 kb upstream and downstream of the predicted genes in silkworms.

The predicted position weight matrices (PWMs) of the recognized sequences of Bm-mamo and the Find Individual Motif Occurrences (FIMO) software of MEME were used to perform silkworm whole-genome scanning for possible downstream target genes. The genomic regions 2 kb upstream and downstream of the 14,623 predicted genes in silkworms were investigated. A total of 10,622 genes contained the recognition sites of the Bm-mamo protein in the silkworm genome (Fig. S9).

Moreover, we compared transcriptome data of the integument tissue between homozygotes and heterozygotes of the *bd* mutant at the 4^{th} instar/beginning molting stage(*52*). In the integument tissue, 10,072 genes (~69% of the total predicted genes of silkworm) were expressed in heterozygotes, and 9,853 genes (~67% of the total predicted genes) were expressed in homozygotes of the *bd* mutant. In addition, there were 191 genes with significant expression differences between homozygotes (*bd*/*bd*) and heterozygotes (+/*bd*) by comparative transcriptome analysis (Table S3)(*52*). Protein functional annotation was performed, and 19 CP genes were significantly differentially expressed between heterozygotes and homozygotes of *bd*. In addition, the orthologs of these CPs were analyzed in *Danaus plexippus*, *Papilio xuthu* and *D. melanogaster*

(Table S4). Furthermore, we identified 53 enzyme-encoding genes, 17 antimicrobial peptide genes, 6 transporter genes, 5 transcription factor genes, 5 cytochrome genes, and others. CP genes were significantly enriched among the differentially expressed genes (DEGs), and previous studies have found that CPs can affect pigmentation(*53*). Therefore, we first conducted an investigation of the CP genes. Among them, 18 CP genes had Bm-mamo binding sites in the upstream and downstream 2 kb genomic regions. In addition, we investigated the expression level of the 18 CP genes in the integument from the 4th instar (day 1) to the beginning of the 5th instar of *Bm-mamo* knockout lines (Fig. S10). The CP genes were significantly upregulated at one or several time points in homozygous (*mamo*$^-$/*mamo*$^-$) individuals compared with heterozygous (*mamo*$^-$/+) *Bm-mamo* knockout line individuals.

Interestingly, the CP gene *BmorCPH24* was significantly upregulated at the feeding stage in *Bm-mamo* knockout homozygotes. Previous studies have shown that BmorCPH24 deficiency can lead to a marked decrease in pigmentation in silkworm larvae(*53*). Therefore, the expression of some CP genes may be necessary for color patterns in caterpillars.

In addition, the synthesis of pigment substances is an important driver of color patterns(*54*). Eight key genes (*TH*, *DDC*, *aaNAT*, *ebony*, *black*, *tan*, *yellow* and *laccase2*) involved in melanin synthesis(*11*) were investigated in heterozygous and homozygous gene knockout lines of *Bm-mamo*. Among them, *yellow*, *tan*, and *DDC* were significantly upregulated during the molting period in the homozygous *Bm-mamo* knockout individuals (**Fig. 6**). The upstream and/or downstream 2 kb genomic regions of *yellow*, *tan* and *DDC* contain the binding site of the Bm-mamo protein. In addition, the expression of *yellow*, *DDC* and *tan* can promote the generation of melanin(*55*).

To explore the interaction between the Bm-mamo protein and its binding sequence, EMSA experiments were conducted. A binding site of Bm-mamo-S (C<u>TGCGT</u>GGT) was located approximately 70 bp upstream of the transcription initiation site of the *Bm-yellow* gene. The EMSA experiment showed that the Bm-mamo-S protein expressed in prokaryotes can bind to the C<u>TGCGT</u>GGT sequence in vitro (Fig. S11). This suggests that the Bm-mamo-S protein can bind to the upstream region of the *Bm-yellow* gene and regulate its transcription.

Therefore, the *Bm-mamo* gene may control the color pattern of caterpillars by regulating key melanin synthesis genes and 18 CP genes.

Discussion

Insects have evolved many important phenotypes during the process of adapting to the environment. Among these traits, color pattern is one of the most interesting. The biochemical metabolic pathways of pigments, such as melanin, ommochromes, and pteridines, have been identified in insects(*18*). However, the regulation of pigment metabolism-related genes and the processes of the transport and deposition of pigment substances are not clear. In this study, we discovered that the *Bm-mamo* gene negatively regulates melanin pigmentation in caterpillars. When this gene is deficient, the body color of silkworms exhibits substantial melanism, and changes in the expression of some melanin synthesis genes and some CP genes are also significant.

Deficiency of some genes in the melanin synthesis pathway, such as *TH*, *yellow*, *ebony*, *tan*, and *aaNAT*, can lead to variations in color patterns (*11*). However, they often lead only to localized pigmentation changes in later-instar caterpillars. For example, in *yellow*-mutant silkworms, the eye spot, lunar spot, star spot, spiracle plate, head, and some sclerotized areas appear reddish brown, and the other cuticle is consistent with that in the wild type in later instars of larvae (third instar to fifth instar)(*56*). This situation is highly similar to that of the *tan* mutant in silkworms(*57*). Why do the *yellow* and *tan* phenotypes appear only in a limited cuticle region in

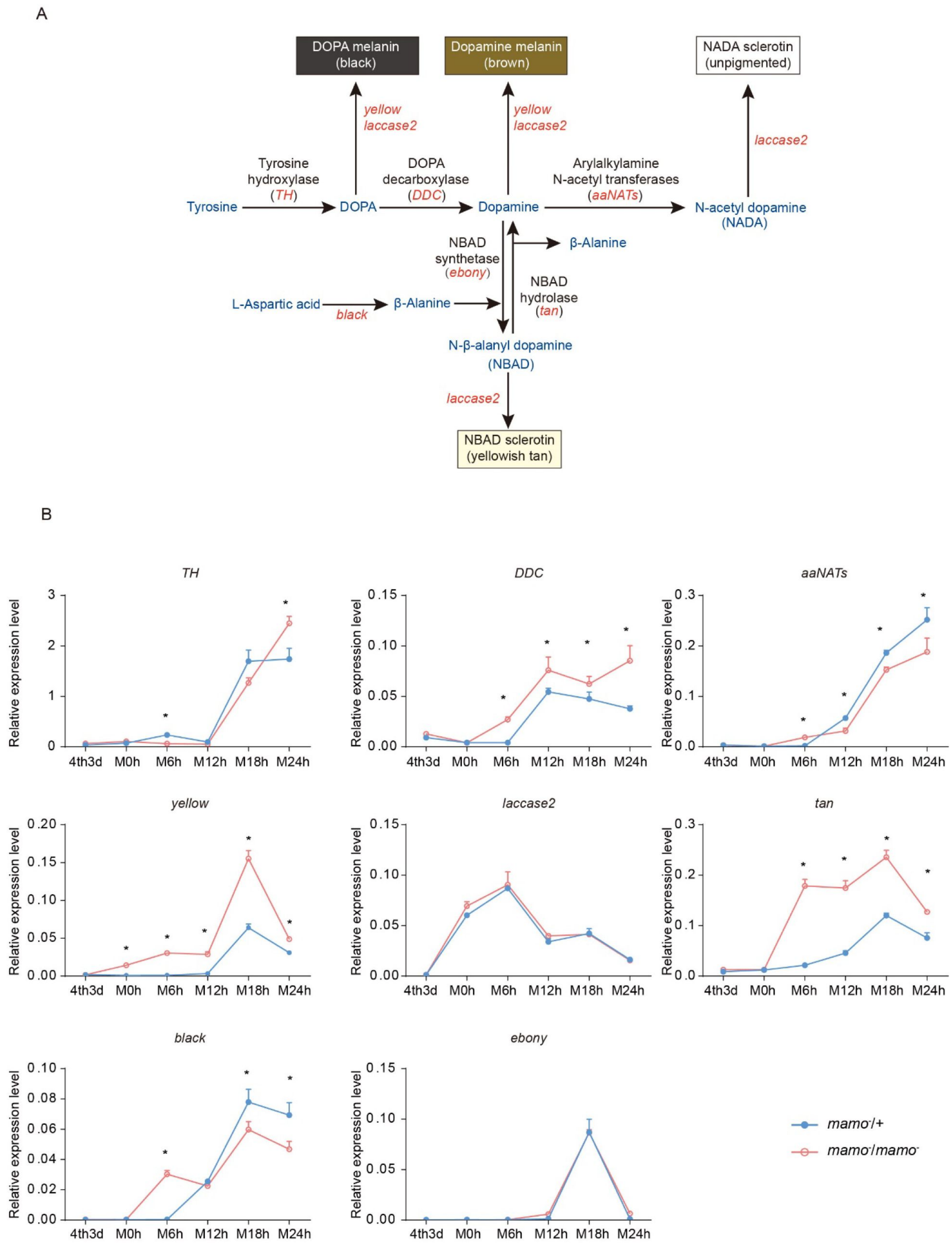

Fig. 6.

Melanin metabolism pathway and qPCR of related genes.

(A) The melanin metabolism pathway. Blue indicates amino acids and catecholamines, red indicates the names of genes, and black indicates the names of enzymes.

(B) Relative expression levels of eight genes in the heterozygous *Bm-mamo* knockout group (blue) and homozygous *Bm-mamo* knockout group (red) as determined by qPCR analysis. The means ± s.d.s. *P<0.05, paired Student's t test. 4th3d indicates 4th instar day 3, M indicates molting, and h indicates hours.

later instars of silkworm larvae? One possible reason is that the expression of *yellow* and *tan* is limited to a certain region by transcription regulators. Alternatively, other factors, such as CPs in the cuticle, may limit pigmentation by interacting with pigment substances in the cuticle.

On the one hand, we investigated the expression levels of *yellow* and *tan* in the pigmented region (lunar spot) and nonpigmented region of the epidermis during the 4th molting of the wild-type Dazao strain. The expression level of *yellow* was significantly upregulated in the lunar spot (the epidermis on the dorsum of the fifth body segment) compared with the nonpigmented region (the epidermis on the dorsum of the sixth body segment) at 6 hours and 12 hours after molting.

Meanwhile, *tan* was significantly upregulated 18 hours after molting in the lunar spot region (Fig. S12). This suggests that the upregulated expression of pigment synthesis genes at key time points may be important for pigmentation. However, *yellow* and *tan* were still moderately expressed in the nonpigmented epidermis, although they did not cause significant melanin pigmentation. This indicates that pigment synthesis alone cannot dominantly determine the color pattern of the cuticle in caterpillars.

On the other hand, synthesized pigment substances need to be transported from epidermal cells and embedded into the cuticle to allow pigmentation. Therefore, the correct cuticle structure and location of cuticular proteins in the cuticle may be important factors affecting pigmentation.

Previous studies have shown that the lack of expression of *BmorCPH24*, which encodes important components of the endocuticle, can lead to dramatic changes in body shape and a significant reduction in the pigmentation of caterpillars(53). We crossed *Bo* (*BmorCPH24* null mutation) and *bd* to obtain F_1(*Bo*/+^{Bo}, *bd*/+), then self-crossed F_1 and observed the phenotype of F_2. The lunar spots and star spots decreased, and light-colored stripes appeared on the body segments, but the other areas still had significant melanin pigmentation in double mutation (*Bo*, *bd*) individuals (Fig. S13). However, in previous studies, introduction of *Bo* into *L* (ectopic expression of *wnt1* results in lunar stripes generated on each body segment)(24) and *U* (overexpression of *SoxD* results in excessive melanin pigmentation of the epidermis)(58) strains by genetic crosses can remarkably reduce the pigmentation of *L* and *U* (53). Interestingly, there was a more significant decrease in pigmentation in the double mutants (*Bo*, *L*) and (*Bo*, *U*) than in (*Bo*, *bd*). This suggests that *Bm-mamo* has a stronger ability than *wnt1* and *SoxD* to regulate pigmentation. On the one hand, mamo may be a stronger regulator of the melanin metabolic pathway, and on the other hand, mamo may regulate other CP genes to reduce the impact of BmorCPH24 deficiency.

How do CPs affect pigmentation? One study showed that some CPs can form “pore canals” to transport some macromolecules(59). In addition, some CPs can be crosslinked with catecholamines, which are synthesized in the melanin metabolism pathway(60). Because there are no live cells in the cuticle, melanin precursor substances may be transported by the pore canals formed by some CPs and fixed to specific positions through cross-linking with CPs. The cuticular protein TcCPR4 is needed for the formation of pore canals of the cuticle in *Tribolium castaneum*(61). In contrast, the vertical pore canal is lacking in the less pigmented cuticles of *T. castaneum*(62). This suggests that the pore canals constructed by TcCPR4 may transport pigments and contribute to cuticle pigmentation in *T. castaneum*. Moreover, the melanin metabolites N-acetyldopamine (NADA) and N-β-alanyldopamine (NBAD) can target and sclerotize the cuticle by cross-linking with specific cuticular proteins (57). This suggests that pigments interact with specific CPs, thereby affecting pigmentation, hardening properties and the structure of the cuticle. Interestingly, a study found that pigments, in addition to absorbing specific wavelengths, can affect cuticle polymerization, density, and refractive index, which in turn affects the reflected wavelengths that produce structural color in butterfly wing scales(63).

This implies that the interaction between pigments and CPs can be very subtle, resulting in the formation of unique nanostructures, such as those on wing scales, that produce brilliant structural colors.

Consequently, to maintain the accuracy of the color pattern, the localization of CPs in the cuticle may be very important. In a previous study employing microarray analysis, different CPs were found in differently colored areas of the epidermis in *Papilio xuthus* larvae(64). We investigated the CP genes highly expressed in the black region of *Papilio xuthus* caterpillars. Thirteen genes had orthologous genes in silkworms (Table S5). Among them, 11 genes (*BmorCPR67*, *BmorCPR71*, *BmorCPR76*, *BmorCPR79*, *BmorCPR99*, *BmorCPR107*, *BmorCPT4*, *BmorCPH5*, *BmorCPFL4*, *BmorCPG27* and *BmorCPG4*) were significantly upregulated in homozygous (*mamo*$^-$/*mamo*$^-$) knockout individuals at some time points from the 3rd day of the 4th instar to the 5th instar; *BmorCPG4* was also among the 18 previously detected CP genes, and two genes (*BmorCPG38* and *BmorCPR129*) were not different between homozygous (*mamo*$^-$/*mamo*$^-$) and heterozygous (*mamo*$^-$/+) individuals (Fig. S14).

The 28 CP genes mentioned above have significantly upregulated expression in homozygous (*mamo*$^-$/*mamo*$^-$) gene knockout individuals at some stages. These CPs may be involved in the transportation or cross-linking of melanin in the cuticle. However, among them, there were no differences in their expression level in some periods compared with the control, and some genes were significantly downregulated at some time points in melanic individuals (**Fig. 7**). This suggests that the regulation of CP genes is complex and may involve other transcription factors and feedback effects. CPs are essential components of the insect cuticle and are involved in cuticular microstructure construction (65), body shape development(66), wing morphogenesis(67), and pigmentation(53). CP genes usually account for over 1% of the total genes in an insect genome and can be categorized into several families, including CPR, CPG, CPH, CPAP1, CPAP3, CPT, CPF and CPFL(68). The CPR family is the largest group of CPs, containing a chitin-binding domain called the Rebers and Riddiford motif (R&R)(69). The variation in the R&R consensus sequence allows subdivision into three subfamilies (RR-1, RR-2, and RR-3)(70). Among the 28 CPs, 11 RR-1 genes, 6 RR-2 genes, 4 hypothetical cuticular protein (CPH) genes, 3 glycine-rich cuticular protein (CPG) genes, 3 cuticular protein Tweedle motif (CPT) genes, and 1 CPFL (like the CPFs in a conserved C-terminal region) gene were identified. The RR-1 consensus among species is usually more variable than RR-2, which suggests that RR-1 may have a species-specific function. RR-2 often clustered into several branches, which may be due to gene duplication events in co-orthologous groups and may result in conserved functions between species (71). The classification of CPH is due to their lack of known motifs. In the epidermis of Lepidoptera, the CPH genes often have high expression levels. For example, *BmorCPH24* had a highest expression level, in silkworm larvae epidermis(72). The CPG protein is rich in glycine. The CPH and CPG genes are less commonly found in insects outside the order Lepidoptera (73). This suggests that they may provide species specific functions for the Lepidoptera. CPT contains a Tweedle motif, and the *TweedleD1* mutation has a dramatic effect on body shape in *D. melanogaster*(74). The CPFL members are relatively conserved in species and may be involved in the synthesis of larval cuticles(75). CPT and CPFL may have relatively conserved functions among insects. The CP genes are a group of rapidly evolving genes, and their copy numbers may undergo significant changes in different species. In addition, RNAi experiments on 135 CP genes in brown planthopper (*Nilaparvata lugens*) showed that deficiency of 32 CP genes leads to significant defective phenotypes, such as lethal, developmental retardation, etc. It is suggested that the 32 CP genes are indispensable, and other CP genes may have redundant and complementary functions (76). In previous studies, it was found that the construction of the larval cuticle of silkworms requires the precise expression of over two hundred CP genes(22). The production, interaction, and deposition of CPs and pigments are complex and precise processes, and our research shows that *Bm-mamo* plays an important regulatory role in this process in silkworm caterpillars. For further understanding of the role of CPs, future work should aim to identify the function of important cuticular protein genes and the deposition mechanism in the cuticle.

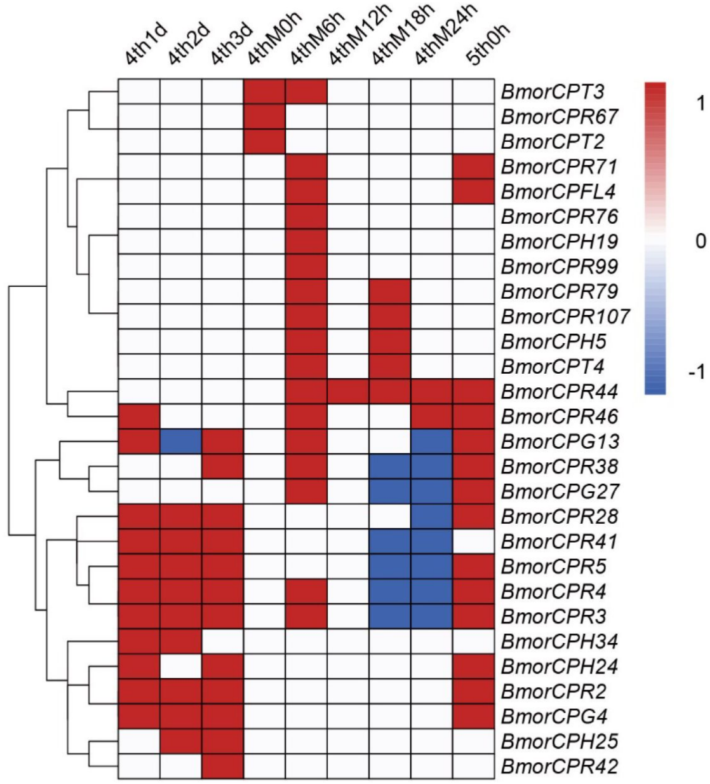

Fig. 7.

Heatmap of 28 differentially expressed cuticular protein genes.

Differentially expressed cuticular protein genes between homozygotes (*mamo*$^-$/*mamo*$^-$) and heterozygotes (*mamo*$^-$/+) of *Bm-mamo* knockout line individuals. Red indicates upregulation in homozygous individuals. Blue indicates downregulation in homozygous individuals. White indicates no difference in expression level or no investigation. 4th1d indicates 4-instar day one, M indicates molting, and h indicates hours.

In addition, among the 191 DEGs found in the comparative transcriptome data, we also discovered some interesting genes. For example, *BGIBMGA013242*, encoding a major facilitator superfamily protein (MFS), named *BmMFS*, which is responsible for the *cheek and tail spot (cts)* mutant, was significantly upregulated in the *bd* mutant. Deficiency of this gene (*BmMFS*) results in chocolate-colored head and anal plates of the silkworm caterpillar(77). In the ommochrome metabolic pathway, *Bm-re*, encoding an MFS protein, may function in the transportation of some amino acids, such as cysteine or methionine, into pigment granules(78). Therefore, the encoded product of *BmMFS* may participate in pigmentation by promoting the maturation of pigment granules. Moreover, *BGIBMGA013576* and *BGIBMGA013656*, which encode MFS domain-containing proteins belonging to the solute carrier family 22 (SLC22) and solute carrier family 2 (SLC2) families, respectively, were significantly upregulated in *bd*. These two genes may participate in pigmentation in a similar manner to the *BmMFS* gene. In addition, the *red Malpighian tubules* (*red*) gene encodes LysM domain-containing proteins. Deficiency of *red* results in a significant decrease in orange pterin in the wing of *Colias* butterflies. Research suggests that the product of *red* may interact with V-ATPase to modulate vacuolar pH across a variety of endosomal organelles, thereby affecting the maturation of pigment granules in cells(79). This indicates that the maturation of pigment granules plays an important role in the coloring of Lepidoptera, and *Bm-mamo* may be involved in regulating the maturation of pigment granules in the epidermal cells of silkworms.

Bm-mamo may affect the synthesis of melanin in epidermis cells by regulating *yellow*, *DDC*, and *tan*; regulate the maturation of melanin granules in epidermis cells through *BmMFS*; and affect the deposition of melanin granules in the cuticle by regulating CP genes, thereby comprehensively regulating the color pattern in caterpillars.

Moreover, in *D. melanogaster*, *mamo* is needed for functional gamete production. In the silkworm, *Bm-mamo* has a conserved function in female reproduction. However, *mamo* has developed new functions in color pattern regulation in silkworm caterpillars. We found that in *D. melanogaster*, the *mamo* gene is mainly expressed during the adult stage; it is not expressed during the 1st instar or 2nd instar larval stage and has very low expression at the 3rd instar (Table S6). In addition, several binding sites of mamo were found near the TTS of *yellow* in *D. melanogaster* (Fig. S15). The *yellow* gene is a key melanin metabolic gene with upstream and intron sequences that have been identified as multiple CREs, and it has been considered a research model for CREs (*80*, *81*). This may be due to a change in the expression pattern of *mamo,* causing this gene to develop a new function of regulating coloration in silkworm caterpillars.

It is generally believed that changes in gene expression patterns are the result of the evolution of CREs. Because CREs have important functions in the spatiotemporal expression pattern regulation of genes, many are found in the noncoding region, which is relatively prone to sequence variation compared with the sequences of coding genes(49). However, the molecular mechanism underlying the sequence change of CREs is not clear.

Transcription factors (TFs), because they recognize relatively short sequences, generally between 4 and 20 bp in length, can have many binding sites in the genome(82). Therefore, one member of the TF family has the potential to regulate almost all genes in one genome. For example, in the PWM sequence of the mamo protein recognition sequence, 10,622 genes (~73% of the total number of genes) contain binding sites within 2 kb of their up/downstream region in silkworms (Table S7 and Table S8). Divergence of CREs is currently considered an important cause of phenotypic evolution(83). For example, the marine form of the three-spined stickleback (*Gasterosteus aculeatus*) has thick armor, and the lake population (which was recently derived from the marine form) does not. Research has shown that pelvic loss in different natural populations of three-spined stickleback fish occurs by regulatory mutations deleting a tissue-specific enhancer (*Pel*) of the *pituitary homeobox transcription factor 1* (*Pitx1*) gene. The researchers genotyped 13 pelvic-reduced populations of three-spined stickleback from disparate geographic locations. Nine of the 13 pelvic-reduced stickleback populations had sequence deletions of varying lengths, all of which were located at the *Pel* enhancer. The author suggested that the *Pitx1* locus of the stickleback genome may be prone to double-stranded DNA breaks that are subsequently repaired by nonhomologous end joining (NHEJ) (84). There are also examples of this in Lepidoptera. The *cortex* gene encodes a member of the Cdc20/fizzy family of cell cycle regulators. The *cortex* gene locus is frequently mutated in Lepidoptera, such as the black (*carbonaria*) form of the peppered moth (*Biston betularia*) in industrial melanism(85), the diversification of the leaf wing of oakleaf butterflies(86), the mimicry wing patterning in *Heliconius* butterflies(87), and the plastic wing pattern in *Junonia coenia* (88). The explanation of DNA fragility is very interesting, but it still cannot explain why DNA sequence mutations occur frequently in a specially site corresponding to the adaptive phenotype. DNA fragility sites are widely distributed in the genome(89). Under the stimulation of the same environmental conditions, all fragile sites are thought to undergo frequent mutations, and this clearly brings adverse burdens to the organism.

Some research suggests that common fragile sites (CFSs) are specific regions in the genome of all individuals that are prone to DNA double stranded breaks (DSBs) and subsequent rearrangements, and the DSB sites are correlated with epigenetic markers for chromatin accessibility, including DNaseI HSSs, H3K4me3, and CTCF binding sites(90, 91).

Moreover, some studies have shown that there is a mutation bias in the genome; compared with the intergenic region, the mutation frequency is reduced by half inside gene bodies and by two-thirds in essential genes. In addition, they compared the mutation rates of genes with different functions. The mutation rate in the coding region of essential genes (such as translation) is the lowest, and the mutation rates in the coding region of specialized functional genes (such as environmental response) are the highest. These patterns are mainly affected by the features of the epigenome(92). Epigenetic modifications include DNA methylation, histone modifications, remodeling of nucleosomes and higher order chromatin reorganization(93). TFs, such as BTB-ZF proteins, can recruit histone modification factors such as DNA methyltransferase 1 (DMNT1), cullin3 (CUL3), histone deacetylase 1 (HDAC1), and histone acetyltransferase 1 (HAT1) to perform chromatin remodeling at specific genomic sites based on specific DNA recognition capabilities(94). In addition, programmed events related to DSBs and repair at genome-specific sites have been reported. For example, PRDM9 can determine recombination hotspots by H3K4 and H3K36 trimethylation (H3K4me3 and H3K36me3) of nucleosomes near its DNA-binding site in meiosis recombination(95). This suggests that, on the basis of randomness, TFs and epigenomic features can determine the mutation bias at unique positions of the genome.

Materials and Methods Silkworm strains

The *bd* and *bd*^{f} mutant strains and wild-type strains Dazao and N4 were obtained from the bank of genetic resources of Southwest University. Silkworms were reared on mulberry leaves or artificial diets at 25 °C and 73% relative humidity under dark for 12 hours and light for 12 hours.

Positional cloning of the *bd* locus

For mapping of the *bd* locus, F1 heterozygous individuals were obtained from a cross between a *bd*^{f} strain and a Dazao strain. Then, an F1 female was crossed with a *bd*^{f} male (BC1F), and an F1 male was backcrossed with a *bd*^{f} female (BC1M). A total of 1162 BC1M individuals were used for recombination analysis. Genomic DNA was extracted from the parents (Dazao, *bd*^{f} and F1) and each BC1 individual using the phenol chloroform extraction method. Available DNA molecular markers were sought through polymorphism screening and linkage analysis. The primers used for mapping are listed in Table S9.

Phylogenetic analysis

To determine whether *Bm-mamo* orthologs existed in other species, the BlastX program of the National Center for Biotechnology Information (NCBI) (*http://www.ncbi.nlm.nih.gov/BLAST/*) was used. The sequences of *Bm-mamo-L* and *Bm-mamo-S* were blasted against the nonredundant protein sequence (nr) database. Sequences with a maximum score and E-value≤10^{-4} were downloaded. The sequences of multiple species were subjected to multiple sequence alignment of the predicted amino acid sequences by MUSCLE(*96*). Then, a phylogenetic tree was constructed using the neighbor-joining method with the MEGA7 program (Pearson model). The confidence levels for various phylogenetic lineages were estimated by bootstrap analysis (2,000 replicates).

Quantitative PCR

Total RNA was isolated from the whole body and integument of the silkworms using TRIzol reagent (Invitrogen, California, USA), purified by phenol chloroform extraction and then reverse transcribed with a PrimeScript™ RT Reagent Kit (TAKARA, Dalian, China) according to the manufacturers' protocol. qPCR was performed using a CFX96™ Real-Time PCR Detection System (Bio-Rad, Hercules, CA) with a Real-Time PCR System and an iTaq Universal SYBR Green Supermix System (Bio-Rad). The cycling parameters were as follows: 95 °C for 3 min, followed by 40 cycles of 95 °C for 10 s and annealing for 30 s. The primers used for target genes are listed in Table S9.

Expression levels for each sample were determined with biological replicates, and each sample was analyzed in triplicate. The gene expression levels were normalized against the expression levels of the ribosomal protein L3 (RpL3). The relative expression levels were analyzed using the classical $R=2^{-\Delta\Delta Ct}$ method.

siRNA for gene knockdown

siRNAs for *Bm-mamo* were designed by the siDirect program (*http://sidirect2.rnai.jp*). The target siRNAs and negative control were synthetized by Tsingke Biotechnology Company Limited. The siRNA (5 μl, 1 μl/μg) was injected from the abdominal spiracle into the hemolymph at the fourth instar (day 3) larval stage. Immediately after injection, phosphate-buffered saline (pH 7.3) droplets were placed nearby, and a 20-voltage pulse for one second and pause for one second were repeated 3 times. The phenotype was observed for fifth-instar larvae. The left and right epidermis were separately dissected from the injected larvae, and RNA was extracted. Then, cDNA was synthetized, and the expression level of the gene was detected by qPCR.

sgRNA synthesis and RNP complex assembly

CRISPRdirect (*http://crispr.dbcls.jp/doc/*) online software was used to screen appropriate single guide RNA (sgRNA) target sequences. The gRNAs were synthetized by Beijing Genomics Institute. sgRNA templates were transcribed using T7 polymerase with RiboMAX™ Large-Scale RNA Production Systems (Promega, Beijing, China) according to the manufacturer's instructions. The RNA transcripts were purified using 3 M sodium acetate (pH 5.2) and anhydrous ethanol (1:30) precipitation, washed using 75% ethanol, and eluted in nuclease-free water. All injection mixes contained 300 ng/μL Cas9 nuclease (Invitrogen, California, USA) and 300 ng/μL purified sgRNA. Before injection, mixtures of Cas9 nuclease and gRNA were incubated for 15 min at 37 °C to reconstitute active ribonucleoproteins (RNPs).

Microinjection of embryos

For embryo microinjection, microcapillary needles were manufactured using a PC-10 model micropipette puller (Narishige, Tokyo, Japan). Microinjection was performed using Eppendorf's TransferMan NK 2 and a FemtoJet 4i system (Eppendorf, Hamburg, Germany). The eggs used for microinjection came from the mating of female and male wild-type Dazao moths. Within 4 hours of being laid, the eggs were adhered onto a clean glass slide. CRISPR/Cas9-messenger RNP mixtures with volumes of approximately 1 nL were injected into the middle of the eggs, and the wound was sealed with glue. All injected embryos were allowed to develop in a climate chamber at 25 °C and 80% humidity.

Comparative genomics

The reference genome of Dazao was downloaded from Silkbase (*https://silkbase.ab.a.u-tokyo.ac.jp/cgi-bin/index.cgi*). The *bd*^{f} genome was obtained from the silkworm pangenome project(34).

The short reads of the *bd*^{f} strains were mapped to the silkworm reference genome by BWA55 v0.7.17 mem with default parameters. The SAMtools56 v1.11 and Picard v2.23.5 (*https://broadinstitute.github.io/picard/*) programs were used to filter the unmapped and duplicated reads. A GVCF file of the samples was obtained using GATK57 v4.1.8.1 HaplotypeCaller with the parameter –ERC = GVCF. The VCF files of insertion/deletions (indels) and single-nucleotide polymorphisms (SNPs) were used for further analysis by eGPS software.

Downstream target gene screening

Online software (*http://zf.princeton.edu*) was used to predict the DNA-binding site for Cys2His2 zinc finger proteins. The confident ZF domains with scores higher than 17.7 were chosen. RF regression on the B1H model was used to predict the DNA-binding sites. Then, the sequence logo and position weight matrices (PWMs) of the DNA-binding sites were obtained. The sequences 2 kb upstream and downstream of the predicted genes were extracted by a Perl script. The FIMO package of the MEME suite was used to search for binding sites in the silkworm genome.

Analysis of EMSA

Primers with binding sites and their flanking sequences were designed according to the FIMO results. A biotin label was added to one end of the upstream primer (probe), and the downstream primer was normal. Primers with the same sequence as the labeled probes were used as competitive probes. The EMSA experiment was conducted with an EMSA reagent kit (Beyotime, Shanghai, China) according to the manufacturers' protocol.

Parthenogenesis

Virgin moths were dissected in 0.9% bacterium-free physiological saline. The ovaries were extracted, and water was removed by clean filter paper. The ovaries were stored at 25 °C and 40% relative humidity for 10 hours. The eggs were treated in 46 °C warm water for 18 minutes and then transferred to room temperature for 5-10 minutes. The activated egg was protected in an environment of 15-17 °C and 18% relative humidity for 3 days. Then, it was immersed in acid immediately (with a hydrochloric acid specific gravity of 1.075 at 46 °C for 5 minutes) or transferred to the same conditions as the hibernating eggs for protection. The average hatching rate of silkworm eggs treated in this way was 50-60%.

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Author contributions

FY Dai conceived and designed the experiments. SY Wu, CX Peng, JW Luo, KP Lu and CH Zhang performed the study. SY Wu, XL Tong, KP Lu, CL Li, X Ding, YR Lu, XH Duan, D Tan and H Hu analyzed the data. SY Wu wrote the paper. XL Tong and FY Dai edited and revised the manuscript.

Competing interests

The authors declare that they have no competing interests.

Data and material availability

The authors confirm that the data supporting the findings of this study are available within the article and its supplementary materials.

Additional information

Supplementary Information is available for this paper.

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Reviewer #1 (Public Review):

Summary: This papers performs fine-mapping of the silkworm mutants bd and its fertile allelic version, bdf, narrowing down the causal intervals to a small interval of a handful of genes. In this region, the gene orthologous to mamo is impaired by a large indel, and its function is later confirmed using expression profiling, RNAi, and CRISPR KO. All these experiments are convincingly showing that mamo is necessary for the suppression of melanic pigmentation in the silkworm larval integument.

The authors also use in silico and in vitro assays to probe the potential effector genes that mamo may regulate.

Strengths: The genotype-to-phenotype workflow, combining forward (mapping) and reverse genetics (RNAi and CRISPR loss-of-function assays) linking mamo to pigmentation are extremely convincing.

This revision is a much improved manuscript and I command the authors for many of their edits.

I find the last part of the discussion, starting at "It is generally believed that changes in gene expression patterns are the result of the evolution of CREs", to be confusing.
In this section, I believe the authors sequentially:
- emphasize the role of CRE in morphological evolution (I agree)
- emphasize that TF, and in particular their own CRE, are themselves important mutational targets of evolution (I agree, but the phrasing need to insist the authors are here talking about the CRE found at the TF locus, not the CRE bound by the TF).
- use the stickleback Pel enhancer as an example, which I think is a good case study, but the authors also then make an argument about DNA fragility sites, which is hard to connect with the present study.

- then continue on "DNA fragility" using the peppered moth and butterfly cortex locus. There is no evidence of DNA fragility at these loci, so the connection does not work. "The cortex gene locus is frequently mutated in Lepidoptera", the authors say. But a more accurate picture would be that the cortex locus is repeatedly involved in the generation of color pattern variants. Unlike for Pel fragile enhancer, we don't know if the causal mutations at this locus are repeatedly the same, and the haplotypes that have been described could be collateral rather than causal. Overall, it is important to clarify the idea that mutation bias is a possible factor explaining "genetic hotspots of evolution" (or genetic parallelism sensu 10.1038/nrg3483), but it is also possible that many genetic hotspots are repeated mutational targets because of their "optimal pleiotropy" (e.g. hub position in GRNs, such as mamo might be), or because of particularly modular CRE region that allow fine-tuning. Thus, I find the "fragility" argument misleading here. In fact the finding that "bd" and "bdf" alleles are different in nature is against the idea of a fragility bias (unless the authors can show increased mutation rates at this locus in a wild silkmoth species?). These alleles are also artificially-selected ie. they increased in frequency by breeding rather than natural selection in the wild, so while interesting for our understand of the genotype-phenotype map, they are not necessarily representative of the mutations that may underlie evolution in the wild.
- Curiously, the last paragraph ("Some research suggests that common fragile sites...") elaborate on the idea that some sites of the genome are prone to mutation. The connection with mamo and the current article are extremely thin. There is here an attempt to connect meiotic and mitotic breaks to Bm-mamo, but this is confusing : it seems to propose Bm-mamo as a recruiter of epigenetic modulators that may drive higher mutation rates elsewhere. Not only I am not convinced by this argument without actual data, but this would not explain how the mutations at the Bm-mamo itself evolved.

On a more positive note, I find it fascinating that the authors identified a TF that clearly articulates or orchestrate larval pattern development, and that when it is deleted, can generate healthy individuals. In other words, while it is a TF with many targets, it is not too pleiotropic. This idea, that the genetically causal modulators of developmental evolution are regulatory genes, has been described elsewhere (e.g. Fig 4c in 10.1038/s41576-020-0234-z, and associated refs). To me, the beautiful findings about Bm-mamo make sense in the general, existing framework that developmental processes and regulatory networks "shape" the evolutionary potential and trajectories of organisms. There is a degree of "programmability" in the genomes, because some loci are particularly prone to modulate a given type of trait. Here, Bm-mamo, as a potentially regulator of both CPs and melanin pathway genes, appear to be a potent modulator of epithelial traits. Claiming that there are inherent mutational biases behind this is unwarranted.

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Author Response

The following is the authors’ response to the original reviews.

> ***Reviewer #1:***
>
> *Summary:*
>
> *This paper performs fine-mapping of the silkworm mutants bd and its fertile allelic version, bdf, narrowing down the causal intervals to a small interval of a handful of genes. In this region, the gene orthologous to mamo is impaired by a large indel, and its function is later confirmed using expression profiling, RNAi, and CRISPR KO. All these experiments are convincingly showing that mamo is necessary for the suppression of melanic pigmentation in the silkworm larval integument. The authors also use in silico and in vitro assays to probe the potential effector genes that mamo may regulate.*

> *Strengths: The genotype-to-phenotype workflow, combining forward (mapping) and reverse genetics (RNAi and CRISPR loss-of-function assays) linking mamo to pigmentation are extremely convincing.*

Response: Thank you very much for your affirmation of our work. The reviewer discussed the parts of our manuscript that involve evolution sentence by sentence. We have further refined the description in this regard and improved the logical flow. Thank you again for your help.

> *Weaknesses:*
>
> 1. *The last section of the results, entitled "Downstream target gene analysis" is primarily based on in silico genome-wide binding motif predictions.*
>
>   *While the authors identify a potential binding site using EMSA, it is unclear how much this general approach over-predicted potential targets. While I think this work is interesting, its potential caveats are not mentioned. In fact the Discussion section seems to trust the high number of target genes as a reliable result. Specifically, the authors correctly say: "even if there are some transcription factor-binding sites in a gene, the gene is not necessarily regulated by these factors in a specific tissue and period", but then propose a biological explanation that not all binding sites are relevant to expression control. This makes a radical short-cut that predicted binding sites are actual in vivo binding sites. This may not be true, as I'd expect that only a subset of binding motifs predicted by Positional Weight Matrices (PWM) are real in vivo binding sites with a ChIP-seq or Cut-and-Run signal. This is particularly problematic for PWM that feature only 5-nt signature motifs, as inferred here for mamo-S and mamo-L, simply because we can expect many predicted sites by chance.*

Response: Thank you very much for your careful work. The analysis and identification of transcription factor-binding sites is an important issue in gene regulation research. Techniques such as ChIP-seq can be used to experimentally identify the binding sites of transcription factors (TFs). However, reports using these techniques often only detect specific cell types and developmental stages, resulting in a limited number of downstream target genes for some TFs. Interestingly, TFs may regulate different downstream target genes in different cell types and developmental stages.

Previous research has suggested that the ZF-DNA binding interface can be understood as a “canonical binding model”, in which each finger contacts DNA in an antiparallel manner. The binding sequence of the C2H2-ZF motif is determined by the amino acid residue sequence of its α-helical component. Considering the first amino acid residue in the α-helical region of the C2H2-ZF domain as position 1, positions -1, 2, 3, and 6 are key amino acids for recognizing and binding DNA. The residues at positions -1, 3, and 6 specifically interact with base 3, base 2, and base 1 of the DNA sense sequence, respectively, while the residue at position 2 interacts with the complementary DNA strand (Wolfe SA et al., 2000; Pabo CO et al., 2001). Based on this principle, the binding sites of C2H2-ZF have good reference value. For the 5-nt PWM sequence, we referred to the study of D. melanogaster, which was identified by EMSA (Shoichi Nakamura et al., 2019). In the new version, we have rewritten this section.

Pabo CO, Peisach E, Grant RA. Design and selection of novel Cys2His2 zinc finger proteins. Annu Rev Biochem. 2001;70:313-340.

Wolfe SA, Nekludova L, Pabo CO. DNA recognition by Cys2His2 zinc finger proteins. Annu Rev Biophys Biomol Struct. 2000;29:183-212.

Nakamura S, Hira S, Fujiwara M, et al. A truncated form of a transcription factor Mamo activates vasa in Drosophila embryos. Commun Biol. 2019;2:422. Published 2019 Nov 20.

> 1. *The last part of the current discussion ("Notably, the industrial melanism event, in a short period of several decades ... a more advanced self-regulation program") is flawed with important logical shortcuts that assign "agency" to the evolutionary process. For instance, this section conveys the idea that phenotypically relevant mutations may not be random. I believe some of this is due to translation issues in English, as I understand that the authors want to express the idea that some parts of the genome are paths of least resistance for evolutionary change (e.g. the regulatory regions of developmental regulators are likely to articulate morphological change). But the language and tone is made worst by the mention that in another system, a mechanism involving photoreception drives adaptive plasticity, making it sound like the authors want to make a Lamarckian argument here (inheritance of acquired characteristics), or a point about orthogenesis (e.g. the idea that the environment may guide non-random mutations).*

Because this last part of the current discussion suffers from confused statements on modes and tempo of regulatory evolution and is rather out of topic, I would suggest removing it.

In any case, it is important to highlight here that while this manuscript is an excellent genotype-to-phenotype study, it has very few comparative insights on the evolutionary process. The finding that mamo is a pattern or pigment regulatory factor is interesting and will deserve many more studies to decipher the full evolutionary study behind this Gene Regulatory Network.

Response: Thank you very much for your careful work. In this part of the manuscript, we introduced some assumptions that make the statement slightly unconventional. The color pattern of insects is an adaptive trait. The bd and bdf mutants used in the study are formed spontaneously. As a frequent variation and readily observable phenotype, color patterns have been used as models for evolutionary research (Wittkopp PJ et al., 2011). Darwin's theory of natural selection has epoch-making significance. I deeply believe in the theory that species strive to evolve through natural selection. However, with the development of molecular genetics, Darwinism’s theory of undirected random mutations and slow accumulation of micromutations resulting in phenotype evolution has been increasingly challenged.

The prerequisite for undirected random mutations and micromutations is excessive reproduction to generate a sufficiently large population. A sufficiently large population can contain sufficient genotypes to face various survival challenges. However, it is difficult to explain how some small groups and species with relatively low fertility rates have survived thus far. More importantly, the theory cannot explain the currently observed genomic mutation bias. In scientific research, every theory is constantly being modified to adapt to current discoveries. The most famous example is the debate over whether light is a particle or a wave, which has lasted for hundreds of years. However, in the 20th century, both sides seemed to compromise with each other, believing that light has a wave−particle duality.

In summary, we have rewritten this section to reduce unnecessary assumptions.

Wittkopp PJ, Kalay G. Cis-regulatory elements: molecular mechanisms and evolutionary processes underlying divergence. Nat Rev Genet. 2011;13(1):59-69.

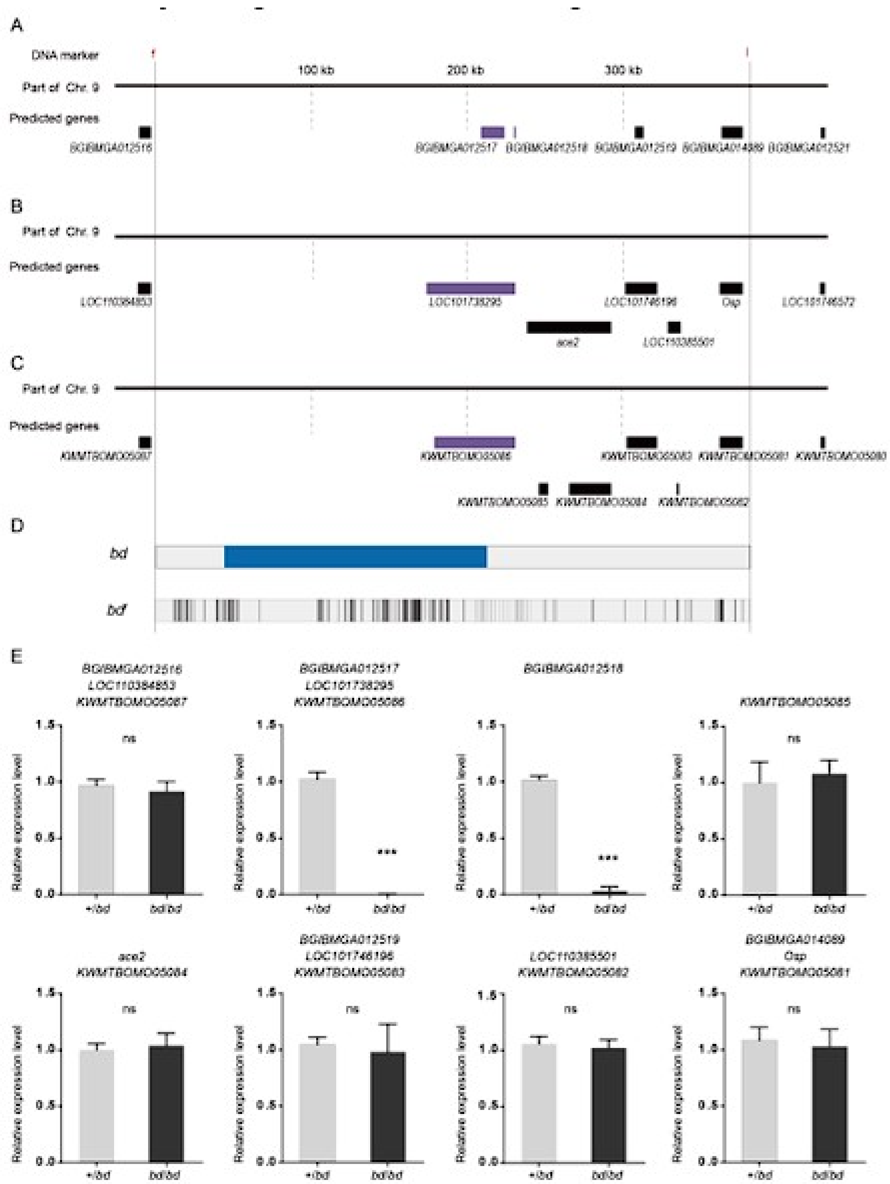
> *Minor Comment:*
>
> *The gene models presented in Figure 1 are obsolete, as there are more recent annotations of the Bm-mamo gene that feature more complete intron-exon structures,*

> *including for the neighboring genes in the bd/bdf intervals. It remains true that the mamo locus encodes two protein isoforms.*
>
> *An example of the Bm-mamo locus annotation, can be found at: https://www.ncbi.nlm.nih.gov/gene/101738295 RNAseq expression tracks (including from larval epidermis) can be displayed in the embedded genome browser from the link above using the "Configure Tracks" tool.*
>
> *Based on these more recent annotations, I would say that most of the work on the two isoforms remains valid, but FigS2, and particularly Fig.S2C, need to be revised.*

Response: Thank you very much for your careful work. In this study, we referred to the predicted genes of SilkDB, NCBI and Silkbase. In different databases, there are varying degrees of differences in the number of predicted genes and the length of gene mRNA. Because the SilkDB database is based on the first silkworm genome, it has been used for the longest time and has a relatively large number of users. In the revised manuscript, we have added the predicted genes of NCBI and Silkbase in Figure S1.

Author response image 1.

The predicted genes and qPCR analysis of candidate genes in the responsible genomic region for bd mutant. (A) The predicted genes in SilkDB;(B) the predicted genes in Genbak;(C) the predicted genes in Silkbase;(D) analysis of nucleotide differences in the responsible region of bd;(E) investigation of the expression level of candidate genes.



Reviewer #2 (Public Review):

Summary:

The authors tried to identify new genes involved in melanin metabolism and its spatial distribution in the silkworm Bombyx mori. They identified the gene Bm-mamo as playing a role in caterpillar pigmentation. By functional genetic and in silico approaches, they identified putative target genes of the Bm-mamo protein. They showed that numerous cuticular proteins are regulated by Bm-mamo during larval development.

Strengths:

- *preliminary data about the role of cuticular proteins to pattern the localization of pigments*
- *timely question*

> - *challenging question because it requires the development of future genetic and cell biology tools at the nanoscale*

Response: Thank you very much for your affirmation of our work. The reviewer's familiarity with the color patterns of Lepidoptera is helpful, and the recommendation raised has provided us with very important assistance. This has allowed us to make significant progress with our manuscript.

> *Weaknesses:*
>
> - *statistical sampling limited*
>
> - *the discussion would gain in being shorter and refocused on a few points, especially the link between cuticular proteins and pigmentation. The article would be better if the last evolutionary-themed section of the discussion is removed.*
>
> *A recent paper has been published on the same gene in Bombyx mori (https://www.sciencedirect.com/science/article/abs/pii/S0965174823000760) in August 2023. The authors must discuss and refer to this published paper through the present manuscript.*

Response: Thank you very much for your careful work. First, we believe that competitive research is sometimes coincidental and sometimes intentional. Our research began in 2009, when we began to configure the recombinant population. In 2016, we published an article on comparative transcriptomics (Wu et al. 2016). The article mentioned above has a strong interest in our research and is based on our transcriptome analysis for further research, with the aim of making a preemptive publication. To discourage such behavior, we cannot cite it and do not want to discuss it in our paper.

Songyuan Wu et al. Comparative analysis of the integument transcriptomes of the black dilute mutant and the wild-type silkworm Bombyx mori. Sci Rep. 2016 May 19:6:26114. doi: 10.1038/srep26114.

> ***Reviewer #1 (Recommendations For The Authors):***
>
> 1. *please consider using a more recent annotation model of the B. mori genome to revise your Result Section 1, Fig.1, and Fig. S2. https://www.ncbi.nlm.nih.gov/gene/101738295*

Specifically, you used BGIM____ gene models, while the current annotation such as the one above featured in the NCBI database provides more accurate intron-exon structures without splitting mamo into tow genes. I believe this can be done with minor revisions of the figures, and you could keep the BGIM____ gene names for the text.

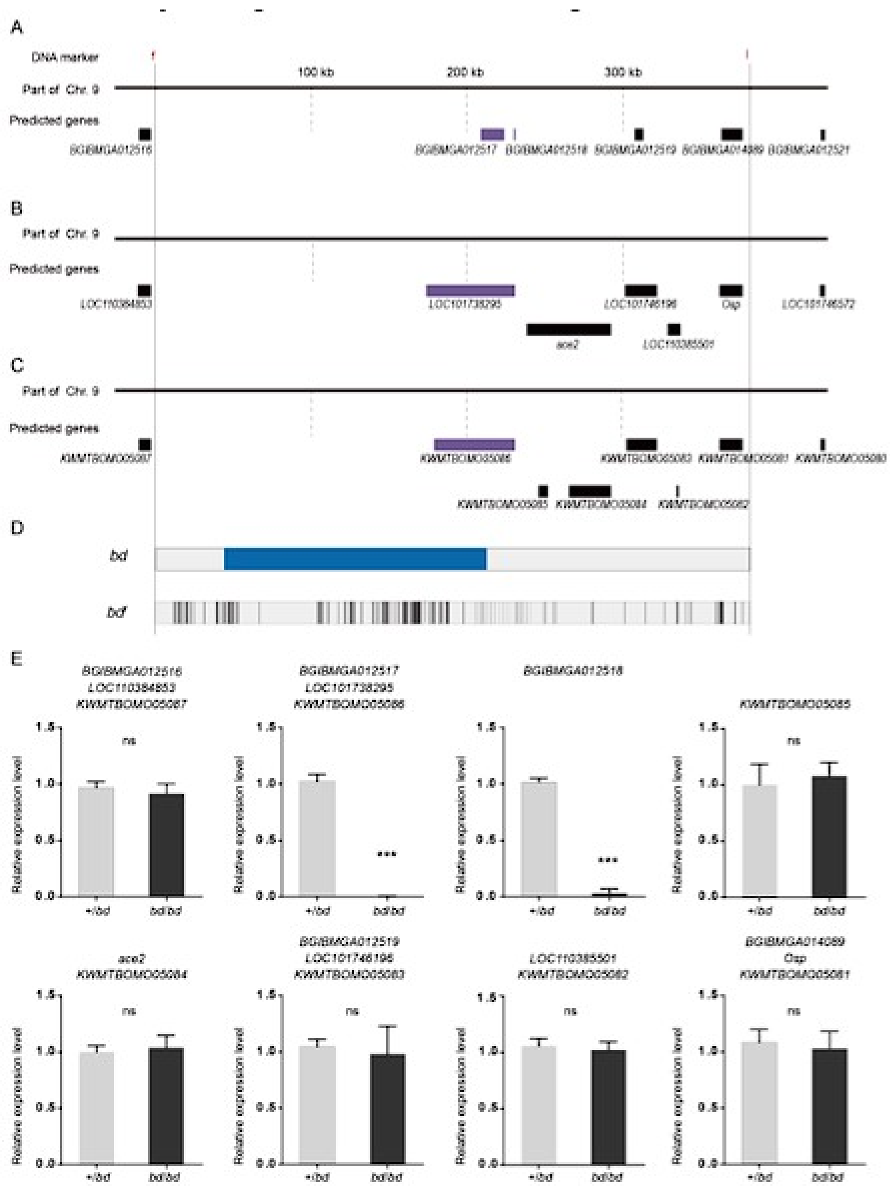
Response: Thank you very much for your careful work. The GenBank of NCBI (National Center for Biotechnology Information) is a very good database that we often use and refer to in this research process. Our research started in 2009, so we mainly referred to the SilkDB database (Jun Duan et al., 2010), although other databases also have references, such as NCBI and Silkbase (https://silkbase.ab.a.u-tokyo.ac.jp/cgi-bin/index.cgi). Because the SilkDB database was constructed based on the first published silkworm genome data, it has been used for the longest time and has a relatively large number of users. Recently, researchers are still using these data (Kejie Li et al., 2023).

The problem with predicting the mamo gene as two genes (BGIBMGA012517 and BGIBMGA012518) in SilkDB is mainly due to the presence of alternative splicing of the mamo gene. BGIBMGA012517 corresponds to the shorter transcript (mamo-s) of the mamo gene. Due to the differences in sequencing individuals, sequencing methods, and methods of gene prediction, there are differences in the number and sequence of predicted genes in different databases. We added the pattern diagram of predicted genes from NCBI and Silkbase, and the expression levels of new predicted genes are shown in Supplemental Figure S1.

Jun Duan et al., SilkDB v2.0: a platform for silkworm (Bombyx mori) genome biology. Nucleic Acids Res. 2010 Jan;38(Database issue): D453-6. doi: 10.1093/nar/gkp801. Kejie Li et al., Transcriptome analysis reveals that knocking out BmNPV iap2 induces apoptosis by inhibiting the oxidative phosphorylation pathway. Int J Biol Macromol. 2023 Apr 1;233:123482. doi: 10.1016/j.ijbiomac.2023.123482. Epub 2023 Jan 31.

Author response image 2.

The predicted genes and qPCR analysis of candidate genes in the responsible genomic region for bd mutant. (A) The predicted genes in SilkDB;(B) the predicted genes in Genbak;(C) the predicted genes in Silkbase;(D) analysis of nucleotide differences in the responsible region of bd;(E) investigation of the expression level of candidate genes.



> 1. *As I mentioned in my public review, I strongly believe the interpretation of the PWM binding analyses require much more conservative statements taking into account the idea that short 5-nt motifs are expected by chance. The work in this section is interesting, but the manuscript would benefit from a quite significant rewrite of the corresponding Discussion section, making it that the in silico approach is prone to the identification of many sites in the genomes, and that very few of those sites are probably relevant for probabilistic reasons. I would recommend statements such as "Future experiments assessing the in vivo binding profile of Bm-mamo (eg. ChIP-seq or Cut&Run), will be required to further understand the GRNs controlled by mamo in various tissues".*

Response: Thank you very much for your careful work. Previous research has suggested that the ZF-DNA binding interface can be understood as a “canonical binding model”, in which each finger contacts DNA in an antiparallel manner. The binding sequence of the C2H2-ZF motif is determined by the amino acid residue sequence of its α-helical component. Considering the first amino acid residue in the α-helical region of the C2H2-ZF domain as position 1, positions -1, 2, 3, and 6 are key amino acids for recognizing and binding DNA. The residues at positions -1, 3, and 6 specifically interact with base 3, base 2, and base 1 of the DNA sense sequence, respectively, while the residue at position 2 interacts with the complementary DNA strand (Wolfe SA et al., 2000; Pabo CO et al., 2001). Based on this principle, the prediction of DNA recognition motifs of C2H2-type zinc finger proteins currently has good accuracy.

The predicted DNA binding sequence (GTGCGTGGC) of the mamo protein in Drosophila melanogaster was highly consistent with that of silkworms. In addition, in D. melanogaster, the predicted DNA binding sequence of mamo, the bases at positions 1 to 7 (GTGCGTG), was highly similar to the DNA binding sequence obtained from EMSA experiments (Seiji Hira et al., 2013). Furthermore, in another study on the mamo protein of Drosophila melanogaster, five bases (TGCGT) were used as the DNA recognition core sequence of the mamo protein (Shoichi Nakamura et al., 2019). In the JASPAR database (https://jaspar.genereg.net), there are also some shorter (4-6 nt) DNA recognition sequences; for example, the DNA binding sequence of Ubx is TAAT (ID MA0094.1) in Drosophila melanogaster. However, we used longer DNA binding motifs (9 nt and 15 nt) of mamo to study the 2 kb genomic regions near the predicted gene. Over 70% of predicted genes were found to have these feature sequences near them. This analysis method is carried out with common software and processes. Due to sufficient target proteins, the accessibility of DNA, the absence of suppressors, the suitability of ion environments, etc., zinc finger protein transcription factors are more likely to bind to specific DNA sequences in vitro than in vivo. Using ChIP-seq or Cut&Run techniques to analyze various tissues and developmental stages in silkworms can yield one comprehensive DNA-binding map of mamo, and some false positives generated by predictions can be excluded. Thank you for your suggestion. We will conduct this work in the next research step. In addition, for brevity, we deleted the predicted data (Supplemental Tables S7 and S8) that used shorter motifs.

Pabo CO, Peisach E, Grant RA. Design and selection of novel Cys2His2 zinc finger proteins. Annu Rev Biochem. 2001;70:313-340.

Wolfe SA, Nekludova L, Pabo CO. DNA recognition by Cys2His2 zinc finger proteins. Annu Rev Biophys Biomol Struct. 2000;29:183-212.

Anton V Persikov et al., De novo prediction of DNA-binding specificities for Cys2His2 zinc finger proteins. Nucleic Acids Res. 2014 Jan;42(1):97-108. doi: 10.1093/nar/gkt890. Epub 2013 Oct 3.

Seiji Hira et al., Binding of Drosophila maternal Mamo protein to chromatin and specific DNA sequences. Biochem Biophys Res Commun. 2013 Aug 16;438(1):156-60. doi: 10.1016/j.bbrc.2013.07.045. Epub 2013 Jul 20.

Shoichi Nakamura et al., A truncated form of a transcription factor Mamo activates vasa in Drosophila embryos. Commun Biol. 2019 Nov 20;2: 422. doi: 10.1038/s42003-019-0663-4. eCollection 2019.

> 1. *In my opinion, the last section of the Discussion needs to be completely removed ("Notably, the industrial melanism event, in a short period of several decades ... a more advanced self-regulation program"), as it is over-extending the data into evolutionary interpretations without any support. I would suggest instead writing a short paragraph asking whether the pigmentary role of mamo is a Lepidoptera novelty, or if it could have been lost in the fly lineage.*
>
> *Below, I tried to comment point-by-point on the main issues I had.*
>
> *Wu et al: Notably, the industrial melanism event, in a short period of several decades, resulted in significant changes in the body color of multiple Lepidoptera species(46). Industrial melanism events, such as changes in the body color of pepper moths, are heritable and caused by genomic mutations(47).*
>
> *Yes, but the selective episode was brief, and the relevant "carbonaria" mutations may have existed for a long time at low-frequency in the population.*

Response: Thank you very much for your careful work. Moth species often have melanic variants at low frequencies outside industrial regions. Recent molecular work on genetics has revealed that the melanic (carbonaria) allele of the peppered moth had a single origin in Britain. Further research indicated that the mutation event causing industrial melanism of peppered moth (Biston betularia) in the UK is the insertion of a transposon element into the first intron of the cortex gene. Interestingly, statistical inference based on the distribution of recombined carbonaria haplotypes indicates that this transposition event occurred in approximately 1819, a date highly consistent with a detectable frequency being achieved in the mid-1840s (Arjen E Van't Hof, et al., 2016). From molecular research, it is suggested that this single origin melanized mutant (carbonaria) was generated near the industrial development period, rather than the ancient genotype, in the UK. We have rewritten this part of the manuscript.

Arjen E Van't Hof, et al., The industrial melanism mutation in British peppered moths is a transposable element. Nature. 2016 Jun 2;534(7605):102-5. doi: 10.1038/nature17951.

Wu et al: If relying solely on random mutations in the genome, which have a time unit of millions of years, to explain the evolution of the phenotype is not enough.

> *What you imply here is problematic for several reasons.*
>
> *First, as you point out later, some large-effect mutations (e.g. transpositions) can happen quickly.*
>
> *Second, it's unclear what "the time units of million of years" means here... mutations occur, segregate in populations, and are selected. The speed of this process depends on the context and genetic architectures.*
>
> *Third, I think I understand what you mean with "to explain the evolution of the phenotype is not enough", but this would probably need a reformulation and I don't think it's relevant to bring it here. After all, you used loss-of-function mutants to explain*

> *the evolution of artificially selected mutants. The evolutionary insights from these mutants are limited. Random mutations at the mamo locus are perfectly sufficient here to explain the bd and bdf phenotypes and larval traits.*

Response: Thank you very much for your careful work. Charles Darwin himself, who argued that “natural selection can act only by taking advantage of slight successive variations; she can never take a leap, but must advance by the shortest and slowest steps” (Darwin, C. R. 1859). This ‘micromutational’ view of adaptation proved extraordinarily influential. However, the accumulation of micromutations is a lengthy process, which requires a very long time to evolve a significant phenotype. This may be only a proportion of the cases. Interestingly, recent molecular biology studies have shown that the evolution of some morphological traits involves a modest number of genetic changes (H Allen Orr. 2005).

One example is the genetic basis analysis of armor-plate reduction and pelvic reduction of the three-spined stickleback (Gasterosteus aculeatus) in postglacial lakes. Although the marine form of this species has thick armor, the lake population (which was recently derived from the marine form) does not. The repeated independent evolution of lake morphology has resulted in reduced armor plate and pelvic structures, and there is no doubt that these morphological changes are adaptive. Research has shown that pelvic loss in different natural populations of three-spined stickleback fish occurs by regulatory mutations deleting a tissue-specific enhancer (Pel) of the pituitary homeobox transcription factor 1 (Pitx1) gene. The researchers genotyped 13 pelvic-reduced populations of three-spined stickleback from disparate geographic locations. Nine of the 13 pelvic-reduced stickleback populations had sequence deletions of varying lengths, all of which were located at the Pel enhancer. Relying solely on random mutations in the genome cannot lead to such similar mutation forms among different populations. The author suggested that the Pitx1 locus of the stickleback genome may be prone to double-stranded DNA breaks that are subsequently repaired by NHEJ (Yingguang Frank Chan et al., 2010).

The bd and bdf mutants used in the study are formed spontaneously. Natural mutation is one of the driving forces of evolution. Nevertheless, we have rewritten the content of this section.

Darwin, C. R. The Origin of Species (J. Murray, London, 1859).

H Allen Orr. The genetic theory of adaptation: a brief history. Nat Rev Genet. 2005 Feb;6(2):119-27. doi: 10.1038/nrg1523.

Yingguang Frank Chan et al., Adaptive evolution of pelvic reduction in sticklebacks by recurrent deletion of a Pitx1 enhancer. Science. 2010 Jan 15;327(5963):302-5. doi: 10.1126/science.1182213. Epub 2009 Dec 10.

> *Wu et al: Interestingly, the larva of peppered moths has multiple visual factors encoded by visual genes, which are conserved in multiple Lepidoptera, in the skin. Even when its compound eyes are covered, it can rely on the skin to feel the color of the environment to change its body color and adapt to the environment(48). Therefore, caterpillars/insects can distinguish the light wave frequency of the background. We suppose that perceptual signals can stimulate the GRN, the GRN guides the expression of some transcription factors and epigenetic factors, and the interaction of epigenetic factors and transcription factors can open or close the chromatin of corresponding downstream genes, which can guide downstream target gene expression.*
>
> *This is extremely confusing because you are bringing in a plastic trait here. It's possible there is a connection between the sensory stimulus and the regulation of mamo in peppered moths, but this is a mere hypothesis. Here, by mentioning a plastic trait, this paragraph sounds as if it was making a statement about directed evolution, especially after implying in the previous sentence that (paraphrasing) "random mutations are not*

> *enough". To be perfectly honest, the current writing could be misinterpreted and co-opted by defenders of the Intelligent Design doctrine. I believe and trust this is not your intention.*

Response: Thank you very much for your careful work. The plasticity of the body color of peppered moth larvae is very interesting, but we mainly wanted to emphasize that their skin shows the products of visual genes that can sense the color of the environment by perceiving light. Moreover, these genes are conserved in many insects. Human skin can also perceive light by opsins, suggesting that they might initiate light–induced signaling pathways (Haltaufderhyde K et al., 2015). This indicates that the perception of environmental light by the skin of animals and the induction of feedback through signaling pathways is a common phenomenon. For clarity, we have rewritten this section of the manuscript.

Haltaufderhyde K, Ozdeslik RN, Wicks NL, Najera JA, Oancea E. Opsin expression in human epidermal skin. Photochem Photobiol. 2015;91(1):117-123.

Wu et al: In addition, during the opening of chromatin, the probability of mutation of exposed genomic DNA sequences will increase (49).

> *Here again, this is veering towards a strongly Lamarckian view with the environment guiding specific mutation. I simply cannot see how this would apply to mamo, nothing in the current article indicates this could be the case here. Among many issues with this, it's unclear how chromatin opening in the larval integument may result in heritable mutations in the germline.*

Response: Thank you very much for your careful work. Previous studies have shown that there is a mutation bias in the genome; compared with the intergenic region, the mutation frequency is reduced by half inside gene bodies and by two-thirds in essential genes. In addition, they compared the mutation rates of genes with different functions. The mutation rate in the coding region of essential genes (such as translation) is the lowest, and the mutation rates in the coding region of specialized functional genes (such as environmental response) are the highest. These patterns are mainly affected by the traits of the epigenome (J Grey Monroe et al., 2022).

In eukaryotes, chromatin is organized as repeating units of nucleosomes, each consisting of a histone octamer and the surrounding DNA. This structure can protect DNA. When one gene is activated, the chromatin region of this gene is locally opened, becoming an accessible region. Research has found that DNA accessibility can lead to a higher mutation rate in the region (Radhakrishnan Sabarinathan et al., 2016; Schuster-Böckler B et al., 2012; Lawrence MS et al., 2013; Polak P et al., 2015). In addition, the BTB-ZF protein mamo belongs to this family and can recruit histone modification factors such as DNA methyltransferase 1 (DMNT1), cullin3 (CUL3), histone deacetylase 1 (HDAC1), and histone acetyltransferase 1 (HAT1) to perform chromatin remodeling at specific genomic sites. Although mutations can be predicted by the characteristics of apparent chromatin, the forms of mutations are diverse and random. Therefore, this does not violate randomness. For clarity, we have rewritten this section of the manuscript.

J Grey Monroe, Mutation bias reflects natural selection in Arabidopsis thaliana. Nature. 2022 Feb;602(7895):101-105.

Sabarinathan R, Mularoni L, Deu-Pons J, Gonzalez-Perez A, López-Bigas N. Nucleotide excision repair is impaired by binding of transcription factors to DNA. Nature. 2016;532(7598):264-267.

Schuster-Böckler B, Lehner B. Chromatin organization is a major influence on regional mutation rates in human cancer cells. Nature. 2012;488(7412):504-507.

Lawrence MS, Stojanov P, Polak P, et al. Mutational heterogeneity in cancer and the search for new cancer-associated genes. Nature. 2013;499(7457):214-218.

Polak P, Karlić R, Koren A, et al. Cell-of-origin chromatin organization shapes the mutational landscape of cancer. Nature. 2015;518(7539):360-364.

Mathew R, Seiler MP, Scanlon ST, et al. BTB-ZF factors recruit the E3 ligase cullin 3 to regulate lymphoid effector programs. Nature. 2012;491(7425):618-621.

> *Wu et al: Transposon insertion occurs in a timely manner upstream of the cortex gene in melanic pepper moths (47), which may be caused by the similar binding of transcription factors and opening of chromatin.*
>
> *No, we do not think that the peppered moth mutation is Lamarckian at all, as seems to be inferred here (notice that by mentioning the peppered moth twice, you are juxtaposing a larval plastic trait and then a purely genetic wing trait, making it even more confusing). Also, the "in a timely manner" is superfluous, because all the data are consistent with a chance mutation being eventually picked up by strong directional mutation. The mutation and selection did NOT occur at the same time.*

Response: Thank you very much for your careful work. The insertion of one transposon into the first intron of the cortex gene of industrial melanism in peppered moth occurred in approximately 1819, which is similar to the time of industrial development in the UK (Arjen E Van't Hof, et al., 2016). In multiple species of Heliconius, the cortex gene is the shared genetic basis for the regulation of wing coloring patterns. Interestingly, the SNP of the cortex, associated with the wing color pattern, does not overlap among different Heliconius species, such as H. erato dephoon and H. erato favorinus, which suggests that the mutations of this cortex gene have different origins (Nadeau NJ et al., 2016). In addition, in Junonia coenia (van der Burg KRL et al., 2020) and Bombyx mori (Ito K et al., 2016), the cortex gene is a candidate for regulating changes in wing coloring patterns. Overall, the cortex gene is an evolutionary hotspot for the variation of multiple butterfly and moth wing coloring patterns. In addition, it was observed that the variations in the cortex are diverse in these species, including SNPs, indels, transposon insertions, inversions, etc. This indicates that although there are evolutionary hotspots in the insect genome, this variation is random. Therefore, this is not completely detached from randomness.

Arjen E Van't Hof, et al., The industrial melanism mutation in British peppered moths is a transposable element. Nature. 2016 Jun 2;534(7605):102-5. doi: 10.1038/nature17951.

Nadeau NJ, Pardo-Diaz C, Whibley A, et al. The gene cortex controls mimicry and crypsis in butterflies and moths. Nature. 2016;534(7605):106-110.

van der Burg KRL, Lewis JJ, Brack BJ, Fandino RA, Mazo-Vargas A, Reed RD. Genomic architecture of a genetically assimilated seasonal color pattern. Science. 2020;370(6517):721-725.

Ito K, Katsuma S, Kuwazaki S, et al. Mapping and recombination analysis of two moth colour mutations, Black moth and Wild wing spot, in the silkworm Bombyx mori. Heredity (Edinb). 2016;116(1):52-59.

> *Wu et al: Therefore, we proposed that the genetic basis of color pattern evolution may mainly be system-guided programmed events that induce mutations in specific genomic regions of key genes rather than just random mutations of the genome.*
>
> *While the mutational target of pigment evolution may involve a handful of developmental regulator genes, you do not have the data to infer such a strong*

> *conclusion at the moment.*
>
> *The current formulation is also quite strong and teleological: "system-guided programmed events" imply intentionality or agency, an idea generally assigned to the anti-scientific Intelligent Design movement. There are a few examples of guided mutations, such as the adaptation phase of gRNA motifs in bacterial CRISPR assays, where I could see the term ""system-guided programmed events" to be applicable. But it is irrelevant here.*

Response: Thank you very much for your careful work. The CRISPR-CAS9 system is indeed very well known. In addition, recent studies have found the existence of a Cas9-like gene editing system in eukaryotes, such as Fanzor. Fanzor (Fz) was reported in 2013 as a eukaryotic TnpB-IS200/IS605 protein encoded by the transposon origin, and it was initially thought that the Fz protein (and prokaryotic TnpBs) might regulate transposon activity through methyltransferase activity (Saito M et al., 2023). Fz has recently been found to be a eukaryotic CRISPR–Cas system. Although this system is found in fungi and mollusks, it raises hopes for scholars to find similar systems in other higher animals. However, before these gene-editing systems became popular, zinc finger nucleases (ZFNs) were already being studied as a gene-editing system in many species. The mechanism by which ZFN recognizes DNA depends on its zinc finger motif (Urnov FD et al., 2005). This is consistent with the mechanism by which transcription factors recognize DNA-binding sites.

Furthermore, a very important evolutionary event in sexual reproduction is chromosome recombination during meiosis, which helps to produce more abundant alleles. Current research has found that this recombination event is not random. In mice and humans, the PRDM9 transcription factors are able to plan the sites of double-stranded breaks (DSBs) in meiosis recombination. PRDM9 is a histone methyltransferase consisting of three main regions: an amino-terminal region resembling the family of synovial sarcoma X (SSX) breakpoint proteins, which contains a Krüppel-associated box (KRAB) domain and an SSX repression domain (SSXRD); a PR/SET domain (a subclass of SET domains), surrounded by a pre-SET zinc knuckle and a post-SET zinc finger; and a long carboxy-terminal C2H2 zinc finger array. In most mammalian species, during early meiotic prophase, PRDM9 can determine recombination hotspots by H3K4 and H3K36 trimethylation (H3K4me3 and H3K36me3) of nucleosomes near its DNA-binding site. Subsequently, meiotic DNA DSBs are formed at hotspots through the combined action of SPO11 and TOPOVIBL. In addition, some proteins (such as RAD51) are involved in repairing the break point. In summary, programmed events of induced and repaired DSBs are widely present in organisms (Bhattacharyya T et al., 2019).

These studies indicate that on the basis of randomness, the genome also exhibits programmability.

Saito M, Xu P, Faure G, et al. Fanzor is a eukaryotic programmable RNA-guided endonuclease. Nature. 2023;620(7974):660-668.

Urnov FD, Miller JC, Lee YL, et al. Highly efficient endogenous human gene correction using designed zinc-finger nucleases. Nature. 2005;435(7042):646-651.

Bhattacharyya T, Walker M, Powers NR, et al. Prdm9 and Meiotic Cohesin Proteins Cooperatively Promote DNA Double-Strand Break Formation in Mammalian Spermatocytes [published correction appears in Curr Biol. 2021 Mar 22;31(6):1351]. Curr Biol. 2019;29(6):1002-1018.e7.

> *Wu et al: Based on this assumption, animals can undergo phenotypic changes more quickly and more accurately to cope with environmental changes. Thus, seemingly complex phenotypes such as cryptic coloring and mimicry that are highly similar to the*

> *background may have formed in a short period. However, the binding sites of some transcription factors widely distributed in the genome may be reserved regulatory interfaces to cope with potential environmental changes. In summary, the regulation of genes is smarter than imagined, and they resemble a more advanced self-regulation program.*
>
> *Here again, I can agree with the idea that certain genetic architectures can evolve quickly, but I cannot support the concept that the genetic changes are guided or accelerated by the environment. And again, none of this is relevant to the current findings about Bm-mamo.*

Response: Thank you very much for your careful work. Darwin's theory of natural selection has epoch-making significance. I deeply believe in the theory that species strive to evolve through natural selection. However, with the development of molecular genetics, Darwinism's theory of undirected random mutations and slow accumulation of micromutations resulting in phenotype evolution has been increasingly challenged.

The prerequisite for undirected random mutations and micromutations is excessive reproduction to generate a sufficiently large population. A sufficiently large population can contain sufficient genotypes to face various survival challenges. However, it is difficult to explain how some small groups and species with relatively low fertility rates have survived thus far. More importantly, the theory cannot explain the currently observed genomic mutation bias. In scientific research, every theory is constantly being modified to adapt to current discoveries. The most famous example is the debate over whether light is a particle or a wave, which has lasted for hundreds of years. However, in the 20th century, both sides seemed to compromise with each other, believing that light has a wave−particle duality.

Epigenetics has developed rapidly since 1987. Epigenetics has been widely accepted, defined as stable inheritance caused by chromosomal conformational changes without altering the DNA sequence, which differs from genetic research on variations in gene sequences. However, an increasing number of studies have found that histone modifications can affect gene sequence variation. In addition, both histones and epigenetic factors are essentially encoded by genes in the genome. Therefore, genetics and epigenetics should be interactive rather than parallel. However, some transcription factors play an important role in epigenetic modifications. Meiotic recombination is a key process that ensures the correct separation of homologous chromosomes through DNA double-stranded break repair mechanisms. The transcription factor PRDM9 can determine recombination hotspots by H3K4 and H3K36 trimethylation (H3K4me3 and H3K36me3) of nucleosomes near its DNA-binding site (Bhattacharyya T et al., 2019). Interestingly, mamo has been identified as an important candidate factor for meiosis hotspot setting in Drosophila (Winbush A et al., 2021).

Bhattacharyya T, Walker M, Powers NR, et al. Prdm9 and Meiotic Cohesin Proteins Cooperatively Promote DNA Double-Strand Break Formation in Mammalian Spermatocytes [published correction appears in Curr Biol. 2021 Mar 22;31(6):1351]. Curr Biol. 2019;29(6):1002-1018.e7.

Winbush A, Singh ND. Genomics of Recombination Rate Variation in Temperature-Evolved Drosophila melanogaster Populations. Genome Biol Evol. 2021;13(1): evaa252.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *Major comments*
>
> - *A recent paper has been published on the same gene in Bombyx mori (https://www.sciencedirect.com/science/article/abs/pii/S0965174823000760) in August*

> *2023. The authors must discuss and refer to this published paper through the present manuscript.*

Response: Thank you very much for your careful work. First, we believe that competitive research is sometimes coincidental and sometimes intentional. Our research began in 2009, when we began to configure the recombinant population. In 2016, we published an article on comparative transcriptomics (Wu et al. 2016). The article mentioned above has a strong interest in our research and is based on our transcriptome analysis for further research, with the aim of making a preemptive publication.

To discourage such behavior, we cannot cite it and do not want to discuss it in our paper.

Songyuan Wu et al. Comparative analysis of the integument transcriptomes of the black dilute mutant and the wild-type silkworm Bombyx mori. Sci Rep. 2016 May 19:6:26114. doi: 10.1038/srep26114.

> - *line 52-54. The numerous biological functions of insect coloration have been thoroughly investigated. It is reasonable to expect more references for each function.*

Response: Thank you very much for your careful work. We have made the appropriate modifications.

Sword GA, Simpson SJ, El Hadi OT, Wilps H. Density-dependent aposematism in the desert locust. Proc Biol Sci. 2000;267(1438):63-68. … Behavior.

Barnes AI, Siva-Jothy MT. Density-dependent prophylaxis in the mealworm beetle Tenebrio molitor L. (Coleoptera: Tenebrionidae): cuticular melanization is an indicator of investment in immunity. Proc Biol Sci. 2000;267(1439):177-182. … Immunity.

N. F. Hadley, A. Savill, T. D. Schultz, Coloration and Its Thermal Consequences in the New-Zealand Tiger Beetle Neocicindela-Perhispida. J Therm Biol. 1992;17, 55-61…. Thermoregulation.

Y. G. Hu, Y. H. Shen, Z. Zhang, G. Q. Shi, Melanin and urate act to prevent ultraviolet damage in the integument of the silkworm, Bombyx mori. Arch Insect Biochem. 2013; 83, 41-55…. UV protection.

M. Stevens, G. D. Ruxton, Linking the evolution and form of warning coloration in nature. P Roy Soc B-Biol Sci. 2012; 279, 417-426…. Aposematism.

K. K. Dasmahapatra et al., Butterfly genome reveals promiscuous exchange of mimicry adaptations among species. Nature.2012; 487, 94-98…. Mimicry.

Gaitonde N, Joshi J, Kunte K. Evolution of ontogenic change in color defenses of swallowtail butterflies. Ecol Evol. 2018;8(19):9751-9763. Published 2018 Sep 3. …Crypsis.

B. S. Tullberg, S. Merilaita, C. Wiklund, Aposematism and crypsis combined as a result of distance dependence: functional versatility of the colour pattern in the swallowtail butterfly larva. P Roy Soc B-Biol Sci.2005; 272, 1315-1321…. Aposematism and crypsis combined.

> - *line 59-60. This general statement needs to be rephrased. I suggest remaining simple by indicating that insect coloration can be pigmentary, structural, or bioluminescent. About the structural coloration and associated nanostructures, the authors could cite recent reviews, such as: Seago et al., Interface 2009 + Lloyd and Nadeau, Current Opinion in Genetics & Development 2021 + "Light as matter:*

> *natural structural colour in art" by Finet C. 2023. I suggest doing the same for recent reviews that cover pigmentary and bioluminescent coloration in insects. The very recent paper by Nishida et al. in Cell Reports 2023 on butterfly wing color made of pigmented liquid is also unique and worth to consider.*

Response: Thank you very much for your careful work. We have made the appropriate modifications.

Insect coloration can be pigmentary, structural, or bioluminescent. Pigments are mainly synthesized by the insects themselves and form solid particles that are deposited in the cuticle of the body surface and the scales of the wings (10, 11). Interestingly, recent studies have found that bile pigments and carotenoid pigments synthesized through biological synthesis are incorporated into body fluids and passed through the wing membranes of two butterflies (Siproeta stelenes and Philaethria diatonica) via hemolymph circulation, providing color in the form of liquid pigments (12). The pigments form colors by selective absorption and/or scattering of light depending on their physical properties (13). However, structural color refers to colors, such as metallic colors and iridescence, generated by optical interference and grating diffraction of the microstructure/nanostructure of the body surface or appendages (such as scales) (14, 15). Pigment color and structural color are widely distributed in insects and can only be observed by the naked eye in illuminated environments. However, some insects, such as fireflies, exhibit colors (green to orange) in the dark due to bioluminescence (16). Bioluminescence occurs when luciferase catalyzes the oxidation of small molecules of luciferin (17). In conclusion, the color patterns of insects have evolved to be highly sophisticated and are closely related to their living environments. For example, cryptic color can deceive animals via high similarity to the surrounding environment. However, the molecular mechanism by which insects form precise color patterns to match their living environment is still unknown.

> - *RNAi approach. I have no doubt that obtaining phenocopies by electroporation might be difficult. However, I find the final sampling a bit limited to draw conclusions from the RT-PCR (n=5 and n=3 for phenocopies and controls). Three control individuals is a very low number. Moreover, it would nice to see the variability on the plot, using for example violin plots.*

Response: Thank you very much for your careful work. In the RNAi experiment, we injected more than 20 individuals in the experimental group and control group. We have added the RNAi data in Figure 4.

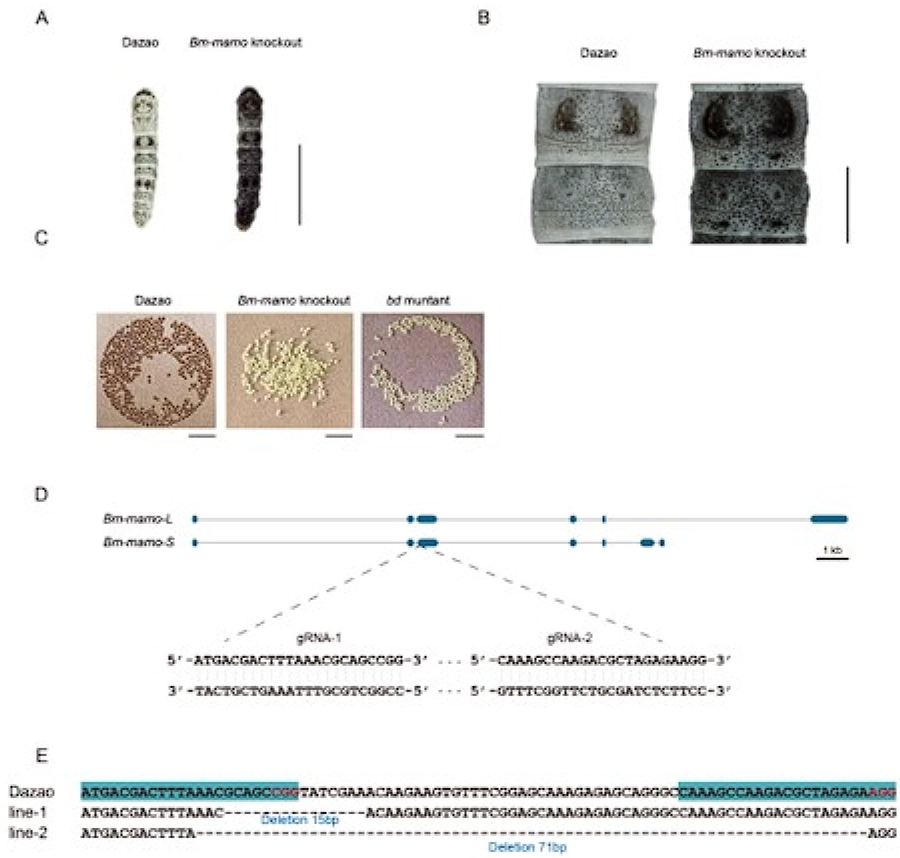
Author response table 1.

	Injected individuals	Melanism phenotype	Survival number	Efficiency
RNAi	48	23	48	47.92%
Control	21	0	17	0

> - *Figure 6. Higher magnification images of Dazao and Bm-mamo knockout are needed, as shown in Figure 5 on RNAi.*

Response: Thank you very much for your careful work. We have added enlarged images.

Author response image 3.



> - *Phylogenetic analysis/Figure S6. I am not sure to what extent the sampling is biased or not, but if not, it is noteworthy that mamo does not show duplicated copies (negative selection?). It might be interesting to discuss this point in the manuscript.*

Response: Thank you very much for your careful work. mamo belongs to the BTB/POZ zinc finger family. The members of this family exhibit significant expansion in vertebrates. For example, there are 3 members in C. elegans, 13 in D. melanogaster, 16 in Bombyx mori, 58 in M. musculus and 63 in H. sapiens (Wu et al, 2019). These members contain conserved BTB/POZ domains but vary in number and amino acid residue compositions of the zinc finger motifs. Due to the zinc finger motifs that bind to different DNA recognition sequences, there may be differences in their downstream target genes. Therefore, when searching for orthologous genes from different species, we required high conservation of their zinc finger motif sequences. Due to these strict conditions, only one orthologous gene was found in these species.

> - *Differentially-expressed genes and CP candidate genes (line 189-191). The manuscript would gain in clarity if the authors explain more in details their procedure. For instance, they moved from a list of 191 genes to CP genes only. Can they say a little bit more about the non-CP genes that are differentially expressed? Maybe quantify the number of CPs among the total number of differentially-expressed genes to show that CPs are the main class?*

Response: Thank you very much for your careful work. The nr (Nonredundant Protein Sequence Database) annotations for 191 differentially expressed genes in Supplemental Table S3 were added. Among them, there were 19 cuticular proteins, 17 antibacterial peptide genes, 6 transporter genes, 5 transcription factor genes, 5 cytochrome genes, 53 enzyme-encoding genes and others. Because CP genes were significantly enriched in differentially expressed genes (DEGs), previous studies have found that BmorCPH24 can affect pigmentation. Therefore, we first conducted an investigation into CP genes.

> - *Interaction between Bm-mamo. It is not clear why the authors chose to investigate the physical interaction of Bm-mamo protein with the putative binding site of yellow, and not with the sites upstream of tan and DDC. Do the authors test one interaction and assume the conclusion stands for the y, tan and DDC?*

Response: Thank you very much for your careful work. In D. melanogaster, the yellow gene is the most studied pigment gene. The upstream and intron sequences of the yellow gene have been identified as containing multiple cis-regulatory elements. Due to the important pigmentation role of the yellow gene and its variable cis-regulatory sequence among different species, it has been considered a research model for cis-regulatory elements (Laurent Arnoult et al. 2013, Gizem Kalay et al. 2019, Yaqun Xin et al. 2020, Yann Le Poul et al. 2020). We use yellow as an example to illustrate the regulation of the mamo gene. We added this description to the discussion.

Laurent Arnoult et al. Emergence and diversification of fly pigmentation through evolution of a gene regulatory module. Science. 2013 Mar 22;339(6126):1423-6. doi: 10.1126/science.1233749.

Gizem Kalay et al. Redundant and Cryptic Enhancer Activities of the Drosophila yellow Gene. Genetics. 2019 May;212(1):343-360. doi: 10.1534/genetics.119.301985. Epub 2019 Mar 6.

Yaqun Xin et al. Enhancer evolutionary co-option through shared chromatin accessibility input. Proc Natl Acad Sci U S A. 2020 Aug 25;117(34):20636-20644. doi: 10.1073/pnas.2004003117. Epub 2020 Aug 10.

Yann Le Poul et al. Regulatory encoding of quantitative variation in spatial activity of a Drosophila enhancer. Sci Adv. 2020 Dec 2;6(49):eabe2955. doi: 10.1126/sciadv.abe2955. Print 2020 Dec.

> - *Please note that some controls are missing for the EMSA experiments. For instance, the putative binding-sites should be mutated and it should be shown that the interaction is lost.*

Response: Thank you very much for your careful work. In this study, we found that the DNA recognition sequence of mamo is highly conserved across multiple species. In D. melanogaster, studies have found that mamo can directly bind to the intron of the vasa gene to activate its expression. The DNA recognition sequence they use is TGCGT (Shoichi Nakamura et al. 2019). We chose a longer sequence, GTGCGTGGC, to detect the binding of mamo. This binding mechanism is consistent across species.

> - *Figure 7 and supplementary data. How did the name of CPs attributed? According to automatic genome annotation of Bm genes and proteins? Based on Drosophila genome and associated gene names? Did the authors perform phylogenetic analyses to name the different CP genes?*

Response: Thank you very much for your careful work. The naming of CPs is based on their conserved motif and their arrangement order on the chromosome. In previous reports, sequence identification and phylogenetic analysis of CPs have been carried out in silkworms (Zhengwen Yan et al. 2022, Ryo Futahashi et al. 2008). The members of the same family have sequence similarity between different species, and their functions may be similar. We have completed the names of these genes in the text, for example, changing CPR2 to BmorCPR2.

Zhengwen Yan et al. A Blueprint of Microstructures and Stage-Specific Transcriptome Dynamics of Cuticle Formation in Bombyx mori. Int J Mol Sci. 2022 May 5;23(9):5155.

Ningjia He et al. Proteomic analysis of cast cuticles from Anopheles gambiae by tandem mass spectrometry. Insect Biochem Mol Biol. 2007 Feb;37(2):135-46.

Maria V Karouzou et al. Drosophila cuticular proteins with the R&R Consensus: annotation and classification with a new tool for discriminating RR-1 and RR-2 sequences. Insect Biochem Mol Biol. 2007 Aug;37(8):754-60.

Ryo Futahashi et al. Genome-wide identification of cuticular protein genes in the silkworm, Bombyx mori. Insect Biochem Mol Biol. 2008 Dec;38(12):1138-46.

> - *Discussion. I think the discussion would gain in being shorter and refocused on the understudied role of CPs. Another non-canonical aspect of the discussion is the reference to additional experiments (e.g., parthogenesis line 290-302, figure S14). This is not the place to introduce more results, and it breaks the flow of the discussion. I encourage the authors to reshuffle the discussion: 1) summary of their findings on mamo and CPs, 2) link between pigmentation mutant phenotypes, pigmentation pattern and CPs, 3) general discussion about the (evo-)devo importance of CPs and link between pigment deposition and coloration. Three important papers should be mentioned here:*
>
> 1. *Matsuoka Y and A Monteiro (2018) Melanin pathway genes regulate color and morphology of butterfly wing scales. Cell Reports 24: 56-65... Yellow has a pleiotropic role in cuticle deposition and pigmentation.*
>
> 1. *https://arxiv.org/abs/2305.16628... Link between nanoscale cuticle density and pigmentation*
>
> 1. *https://www.cell.com/cell-reports/pdf/S2211-1247(23)00831-8.pdf... Variation in pigmentation and implication of endosomal maturation (gene red).*

Response: Thank you very much for your careful work. We have rewritten the discussion section.

1. We have summarized our findings.

Bm-mamo may affect the synthesis of melanin in epidermis cells by regulating yellow, DDC, and tan; regulate the maturation of melanin granules in epidermis cells through BmMFS; and affect the deposition of melanin granules in the cuticle by regulating CP genes, thereby comprehensively regulating the color pattern in caterpillars.

1. We describe the relationship among the pigmentation mutation phenotype, pigmentation pattern, and CP.

Previous studies have shown that the lack of expression of BmorCPH24, which encodes important components of the endocuticle, can lead to dramatic changes in body shape and a significant reduction in the pigmentation of caterpillars (53). We crossed Bo (BmorCPH24 null mutation) and bd to obtain F1(Bo/+Bo, bd/+), then self-crossed F1 and observed the phenotype of F2. The lunar spots and star spots decreased, and light-colored stripes appeared on the body segments, but the other areas still had significant melanin pigmentation in double mutation (Bo, bd) individuals (Fig. S13). However, in previous studies, introduction of Bo into L (ectopic expression of wnt1 results in lunar stripes generated on each body segment) (24) and U (overexpression of SoxD results in excessive melanin pigmentation of the epidermis) (58) strains by genetic crosses can remarkably reduce the pigmentation of L and U (53). Interestingly, there was a more significant decrease in pigmentation in the double mutants (Bo, L) and (Bo, U) than in (Bo, bd). This suggests that Bm-mamo has a stronger ability than wnt1 and SoxD to regulate pigmentation. On the one hand, mamo may be a stronger regulator of the melanin metabolic pathway, and on the other hand, mamo may regulate other CP genes to reduce the impact of BmorCPH24 deficiency.

1. We discussed the importance of (evo-) devo in CPs and the relationship between pigment deposition and coloring.

CP genes usually account for over 1% of the total genes in an insect genome and can be categorized into several families, including CPR, CPG, CPH, CPAP1, CPAP3, CPT, CPF and CPFL (68). The CPR family is the largest group of CPs, containing a chitin-binding domain called the Rebers and Riddiford motif (R&R) (69). The variation in the R&R consensus sequence allows subdivision into three subfamilies (RR-1, RR-2, and RR-3) (70). Among the 28 CPs, 11 RR-1 genes, 6 RR-2 genes, 4 hypothetical cuticular protein (CPH) genes, 3 glycine-rich cuticular protein (CPG) genes, 3 cuticular protein Tweedle motif (CPT) genes, and 1 CPFL (like the CPFs in a conserved C-terminal region) gene were identified. The RR-1 consensus among species is usually more variable than RR-2, which suggests that RR-1 may have a species-specific function. RR-2 often clustered into several branches, which may be due to gene duplication events in co-orthologous groups and may result in conserved functions between species (71). The classification of CPH is due to their lack of known motifs. In the epidermis of Lepidoptera, the CPH genes often have high expression levels. For example, BmorCPH24 had a highest expression level, in silkworm larvae epidermis (72). The CPG protein is rich in glycine. The CPH and CPG genes are less commonly found in insects outside the order Lepidoptera (73). This suggests that they may provide species specific functions for the Lepidoptera. CPT contains a Tweedle motif, and the TweedleD1 mutation has a dramatic effect on body shape in D. melanogaster (74). The CPFL members are relatively conserved in species and may be involved in the synthesis of larval cuticles (75). CPT and CPFL may have relatively conserved functions among insects. The CP genes are a group of rapidly evolving genes, and their copy numbers may undergo significant changes in different species. In addition, RNAi experiments on 135 CP genes in brown planthopper (Nilaparvata lugens) showed that deficiency of 32 CP genes leads to significant defective phenotypes, such as lethal, developmental retardation, etc. It is suggested that the 32 CP genes are indispensable, and other CP genes may have redundant and complementary functions (76). In previous studies, it was found that the construction of the larval cuticle of silkworms requires the precise expression of over two hundred CP genes (22). The production, interaction, and deposition of CPs and pigments are complex and precise processes, and our research shows that Bm-mamo plays an important regulatory role in this process in silkworm caterpillars. For further understanding of the role of CPs, future work should aim to identify the function of important cuticular protein genes and the deposition mechanism in the cuticle.

> *Minor comments*

> - *Title. At this stage, there is no evidence that Bm-mamo regulates caterpillar pigmentation outside of Bombyx mori. I suggest to precise 'silkworm caterpillars' in the title.*

Response: Thank you very much for your careful work. We have modified the title.

> - *Abstract, line 29. Because the knowledge on pigmentation pathway(s) is advanced, I would suggest writing 'color pattern is not fully understood' instead of 'color pattern is not clear'.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 29. I suggest 'the transcription factor' rather than 'a transcription factor'.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 30. If you want to mention the protein, the name 'Bm-mamo' should not be italicized.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 30. 'in the silkworm'.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 31. 'mamo' should not be italicized.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 31. 'in Drosophila' rather 'of Drosophila'.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 32. Bring detail if the gamete function is conserved in insects? In all animals?*

Response: Thank you very much for your careful work. The sentence was changed to "This gene has a conserved function in gamete production in Drosophila and silkworms and evolved a pleiotropic function in the regulation of color patterns in caterpillars."

- Introduction, line 51. I am not sure what the authors mean by 'under natural light'. Please rephrase.

Response: Thank you very much for your careful work. We have deleted "under natural light".

> - *line 43. I find that the sentence 'In some studies, it has been proven that epidermal proteins can affect the body shape and appendage development of insects' is not necessary here. Furthermore, this sentence breaks the flow of the teaser.*

Response: Thank you very much for your careful work. We have deleted this sentence.

> - *line 51-52. 'Greatly benefit them' should be rephrased in a more neutral way. For example, 'colours pattern have been shown to be involved in...'.*

Response: Thank you very much for your careful work. We have modified to "and the color patterns have been shown to be involved in…"

> - *line 62. CPs are secreted by the epidermis, but I would say that CPs play their structural role in the cuticle, not directly in the epidermis. I suggest rephrasing this sentence and adding references.*

Response: Thank you very much for your careful work. We have modified "epidermis" to "cuticle".

> - *line 67. Please indicate that pathways have been identified/reported in Lepidoptera (11). Otherwise, the reader does not understand if you refer to previous biochemical in Drosophila for example.*

Response: Thank you very much for your careful work. We have modified this sentence. "Moreover, the biochemical metabolic pathways of pigments used for color patterning in Lepidoptera…have been reported."

> - *line 69. Missing examples of pleiotropic factors and associated references. For example, I suggest adding: engrailed (Dufour, Koshikawa and Finet, PNAS 2020) + antennapedia (Prakash et al., Cell Reports 2022) + optix (Reed et al., Science 2011), etc. Need to add references for clawless, abdominal-A.*

Response: Thank you very much for your careful work. We have made modifications.

> - *line 76. The simpler term moth might be enough (instead of Lepidoptera).*

Response: Thank you very much for your careful work. We have modified this to "insect".

> - *line 96. I would simplify the text by writing "Then, quantitative RT-PCR was performed..."*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 112. 'Predict' instead of 'estimate'?*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 113. I would rather indicate the full name first, then indicate mamo between brackets.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 144. The Perl script needs to be made accessible on public repository.*

Response: Thank you very much for your careful work.

> - *line 147-150. Too many technical details here. The details are already indicated in the material and methods section. Furthermore, the details break the flow of the paragraph.*

Response: Thank you very much for your careful work. We have modified this section.

> - *line 152. Needs to make the link with the observed phenotypes in Figure 1. Just needs to state that RNAi phenocopies mimic the mutant alleles.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 153-157. Too many technical details here. The details are already indicated in the material and methods section. Furthermore, the details break the flow of the paragraph.*

Response: Thank you very much for your careful work. We have simplified this paragraph.

> - *line 170. Please rephrase 'conserved in 30 species' because it might be understood as conserved in 30 species only, and not in other species.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 182. Maybe explain the rationale behind restricting the analysis to +/- 2kb. Can you cite a paper that shows that most of binding sites are within 2kb from the start codon?*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 182. '14,623 predicted genes'.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 183. '10,622 genes'*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 183. Redundancy. Please remove 'silkworm' or 'B. mori'.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 187. '10,072 genes'*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 188. '9,853 genes'*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 200. "Therefore, the differential...in caterpillars" is a strong statement.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 204. Remove "The" in front of eight key genes. Also, needs a reference... maybe a recent review on the biochemical pathway of melanin in insects.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 220. This sentence is too general and vague. Please explicit what you mean by "in terms of evolution". Number of insect species? Diversity of niche occupancy? Morphological, physiological diversity?*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 285. The verb "believe" should be replaced by a more neutral one.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 354-355. This sentence needs to be rephrased in a more objective way.*

Response: Thank you very much for your careful work. We have rewritten this sentence.

> - *line 378. Missing reference for MUSCLE.*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 379. Pearson model?*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *line 408. "The CRISPRdirect online software was used...".*

Response: Thank you very much for your careful work. We have modified this sentence.

> - *Figure 1. In the title, I suggest indicating Dazao, bd, bdf as it appears in the figure. Needs to precise 'silkworm larval development'.*

Response: Thank you very much for your careful work. We have modified this figure title.

> - *Figure 3. In the title, is the word 'pattern' really necessary? In the legend, please indicate the meaning of the acronyms AMSG and PSG.*

Response: Thank you very much for your careful work. We have modified this figure legend.

> - *Figure S7A. Typo 'Znic finger 1', 'Znic finger 2', 'Znic finger 3',*

Response: Thank you very much for your careful work. We have fixed these typos. .